

Weeks 2-10

Grades 4-5

Investigate all cases, prove an obstruction, or explain an optimum.

Week 2 is one shared collection; grade bands below apply to Weeks 3-10.

Give one page at a time. The packet is a menu, not a checklist for one hour.

Each student set includes the same Week 2 library. Print its chosen pages once.

Week and investigation	PDF pages
02 Switches and lamps	2-43
03 Repeating secret machines	44-47
04 Around the ring	48-51
05 Tower cities	52-56
06 Code-breaking with guarantees	57-60
07 Take-away games	61-64
08 Rooks and piles	65-69
09 Bouncing paths	70-74
10 Bridges and delivery routes	75-79

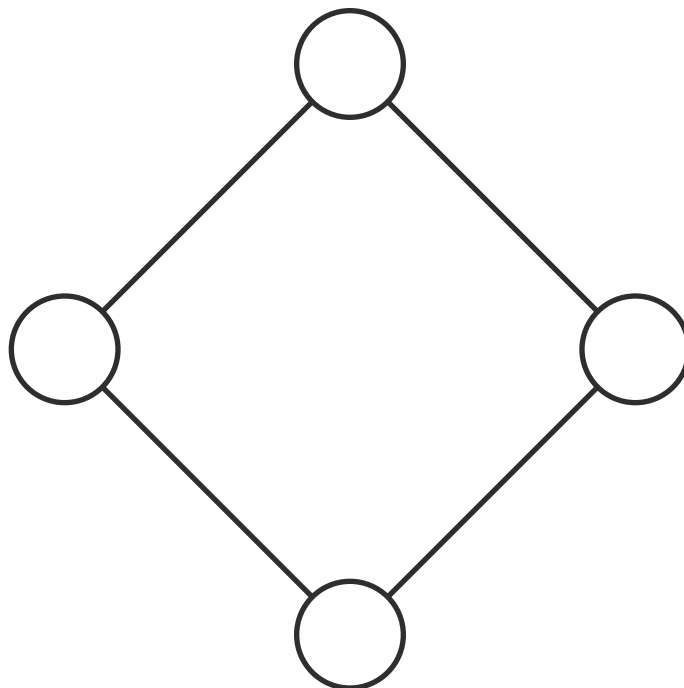
Print US Letter, single-sided, at 100% / Actual Size.

Page ranges above count this cover as page 1. Bookmarks open each week.

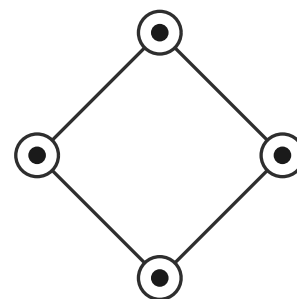
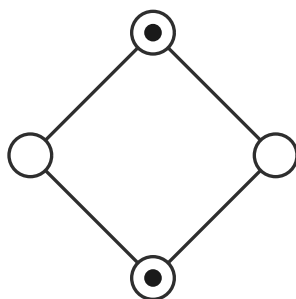
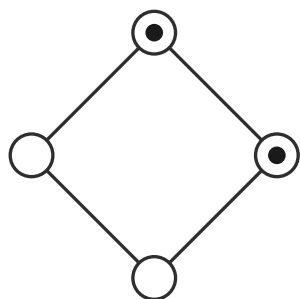
Weekly footers retain their own page numbers and activity IDs for the use log.

Unpiloted. Week 2 updated September 28; other core revisions September 27.

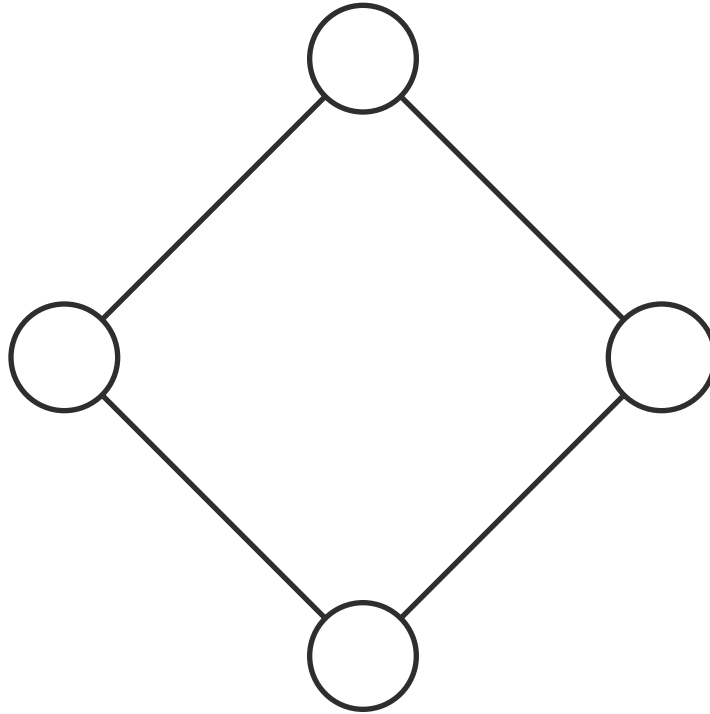
Problem 1: Empty is OFF. A dot is ON. Choose a line. Change BOTH lamps at its ends: add a dot or erase a dot. Try a move. Try that same move again.



Problem 2: Start all OFF each time. Make each small picture on the big board.

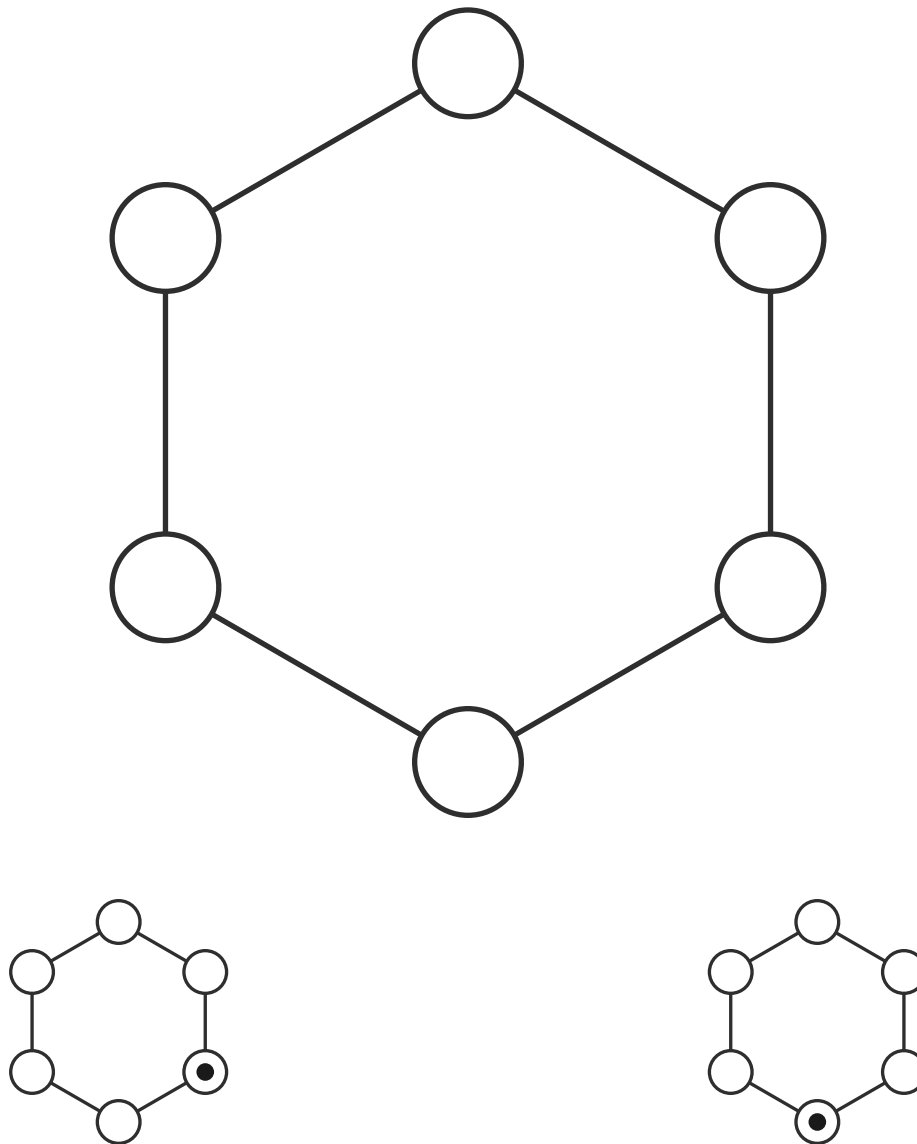


Problem 3: Play with a partner. Start all OFF. Make one move: change both ends of a line. Can your partner turn all the lamps OFF? Swap jobs.



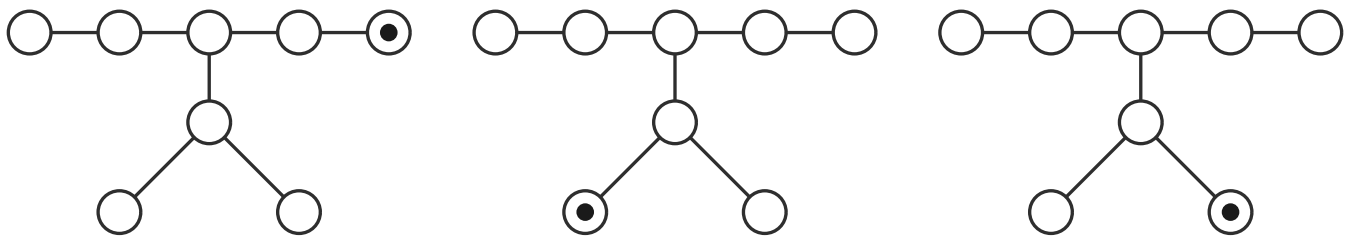
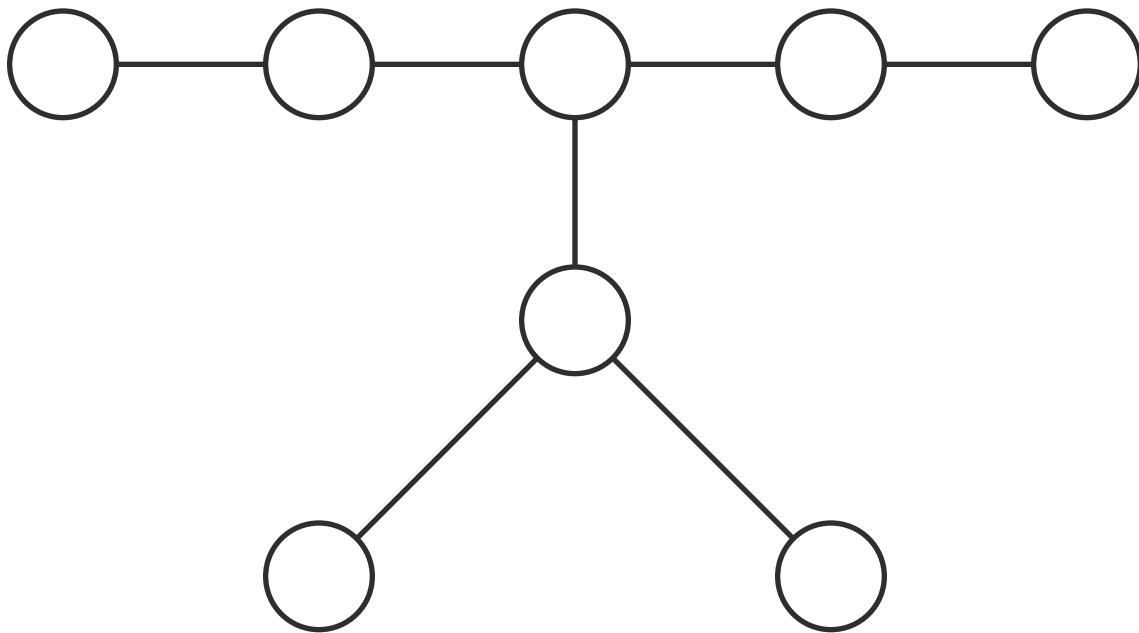
Problem 4: Start all OFF. Make TWO moves for your partner to undo. Swap jobs. Then try THREE moves. Show your moves if your partner wants a hint.

Problem 5: Start with only the top lamp ON each time. Change both ends of a line to make each small picture. Keep exactly ONE lamp ON after every move.

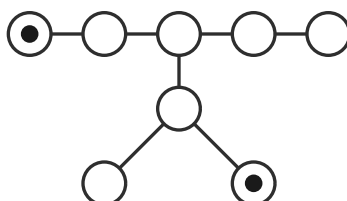


Problem 6: Start at the top again. Visit every lamp and bring the light home.

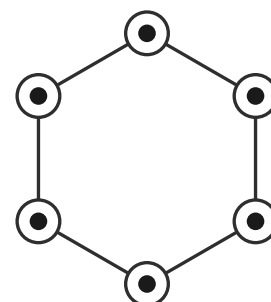
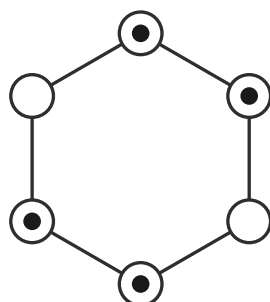
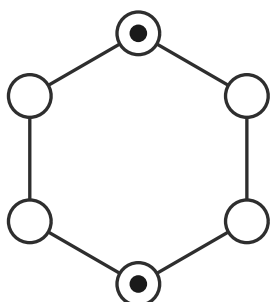
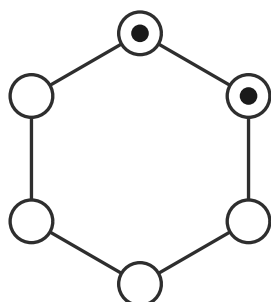
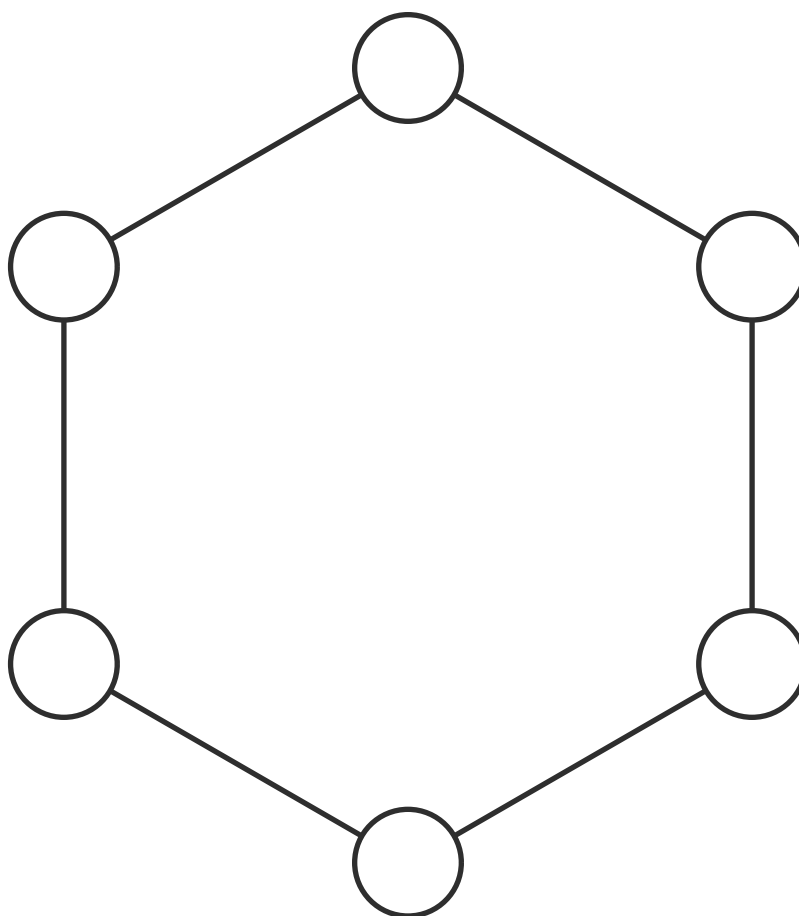
Problem 7: Start with only the far-left lamp ON each time. Change both ends of a line. Move the light to match each small picture. Keep ONE lamp ON.



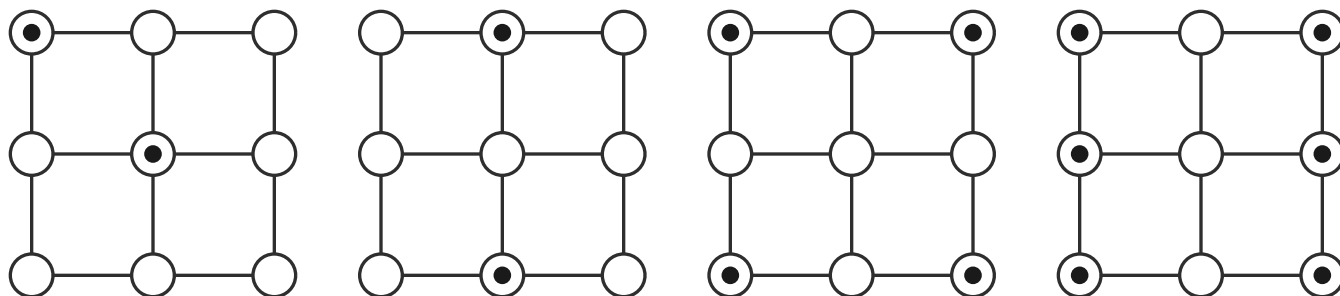
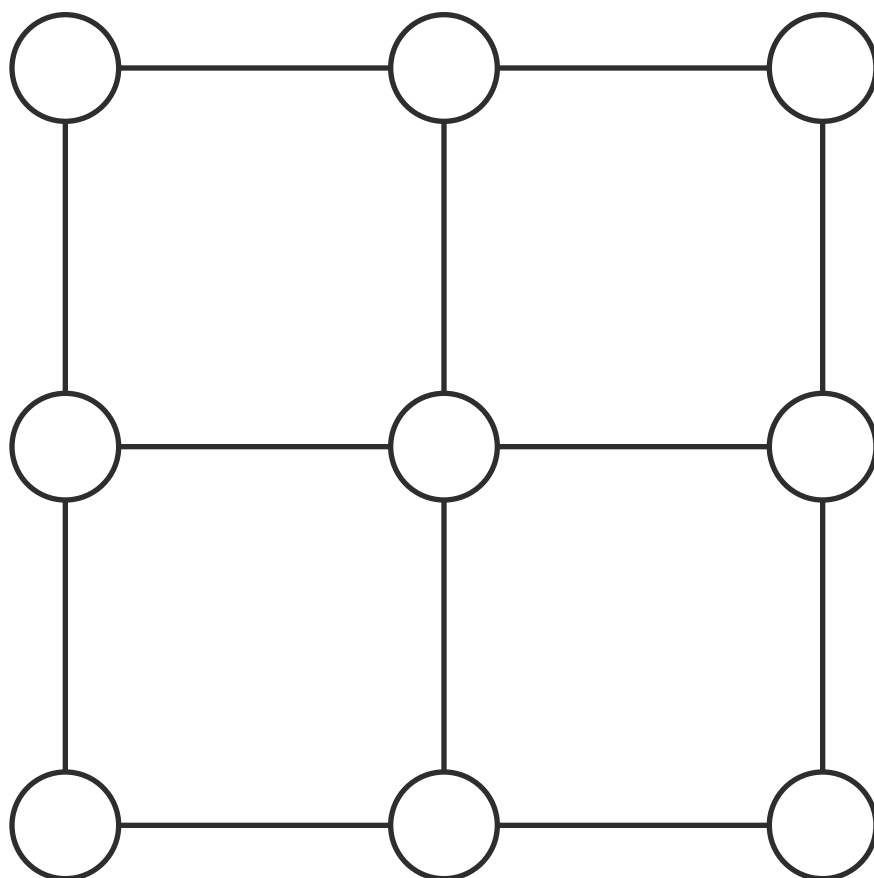
Problem 8: Now start all OFF. Make this two-light picture. You may have more than one light ON.



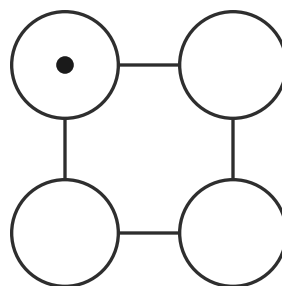
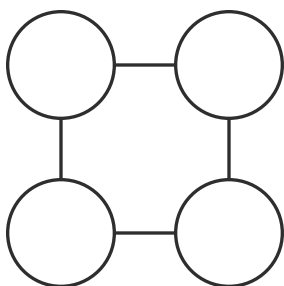
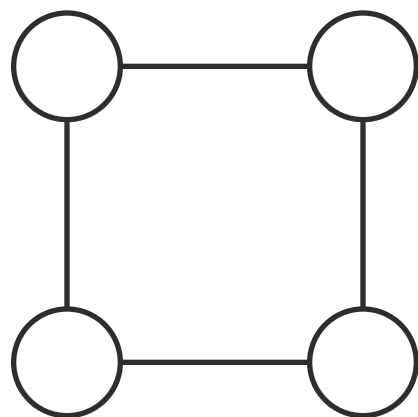
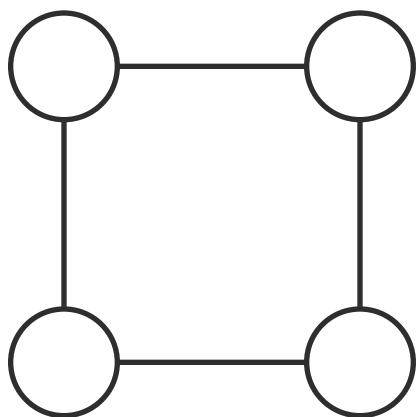
Problem 9: Start all OFF for each small picture. Change BOTH ends of one line each move. Make the picture on the big board. Point to one you want to do again.



Problem 10: Start all OFF for each small picture. Change BOTH ends of one line each move. Make the picture on the big board. Try a different way to make one picture.



Problem 11: Put a dot in the top-left lamp of the LEFT island. Change BOTH ends of one line each move. Can you make the small picture?



Problem 12: Draw ONE new line between lamps on different islands. Start again with only the top-left lamp of the left island ON. Try the trip with your bridge.

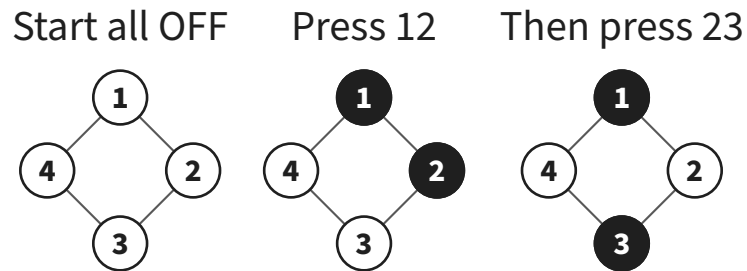
Problem 13: Draw 4 to 8 lamps. Join them with lines so every lamp has a path to every other lamp. Keep lines from crossing. Start all OFF. Make TWO moves, changing both ends of a line each time. Let a partner turn them all OFF.



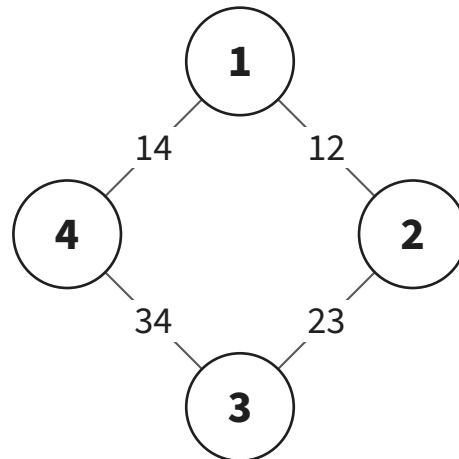
Problem 14: Add one new lamp. Join it to an old lamp with a line. Make a new puzzle for your partner.

Problem 15: Use the line names to save your moves. Start all OFF on the big board. Press a line: change *both* end lamps. Add or erase pencil dots, or flip two-sided counters. Dotted or dark means ON.

The line joining lamps 1 and 2 is called **12**. Replay these two moves on the big board.



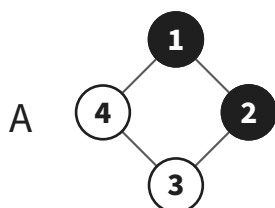
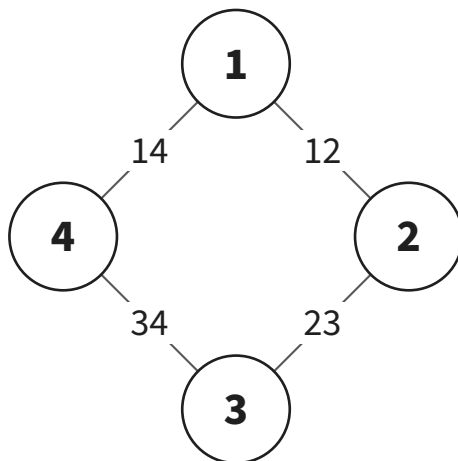
Save the moves in order: **12, 23**. You may ask someone to write while you point.



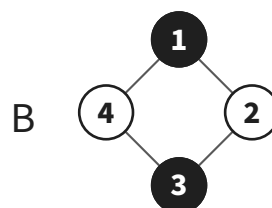
Get back to all OFF by pressing lines. Save your moves so a partner can replay them.

Moves: _____

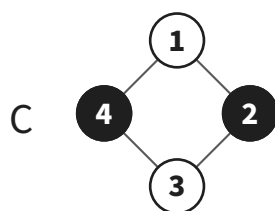
Problem 16: Make each target picture on the big board. Reset all lamps to OFF before each target. Change both ends of one line at a time. Write your moves in order.



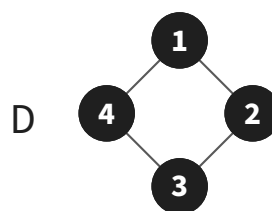
Moves: _____



Moves: _____



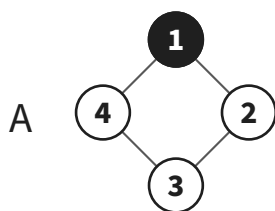
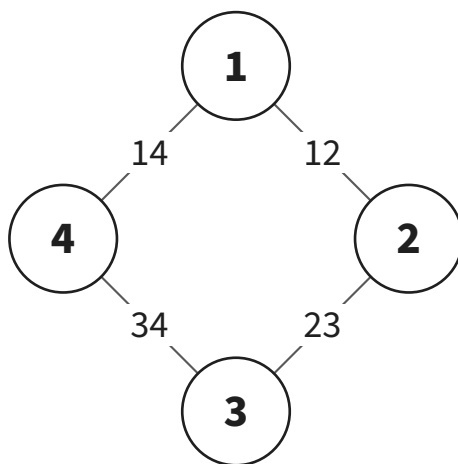
Moves: _____



Moves: _____

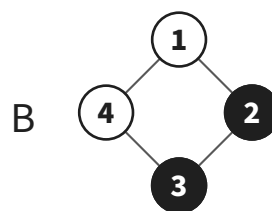
Give a partner one of your move lists. Reset all OFF. Have your partner replay it and point to its target.

Problem 17: Try each target picture. Start all OFF each time. Change both ends of one line at a time. Write a move list that works, or check “not yet.”



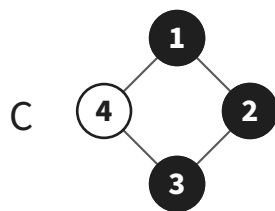
Moves: _____

☐ Not yet



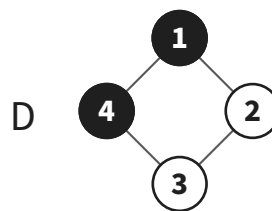
Moves: _____

☐ Not yet



Moves: _____

☐ Not yet

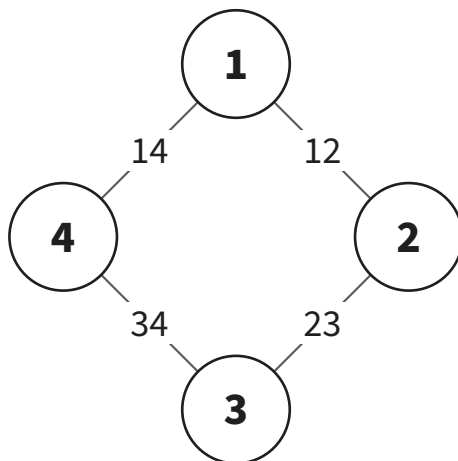


Moves: _____

☐ Not yet

For a “not yet” picture, try a different move list on the board. Keep any list that works.

Problem 18: For each row, set up its starting picture by hand on the big board: add dots or turn counters over. Leave lamps 3 and 4 OFF. Then press 12 once. Darken the ON lamps in the answer picture. Count ON lamps before and after.



A  — 12 — 
ON before: _____

press 12 →

 — 12 — 
ON after: _____

B  — 12 — 
ON before: _____

press 12 →

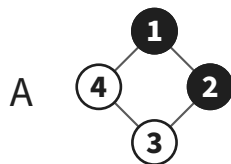
 — 12 — 
ON after: _____

C  — 12 — 
ON before: _____

press 12 →

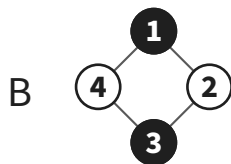
 — 12 — 
ON after: _____

Problem 19: These are the targets you tried. For each picture, circle “made” or “not yet.” Count its ON lamps. Use the big board to try any picture again; reset all OFF each time.



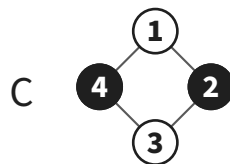
ON: _____

made / not yet



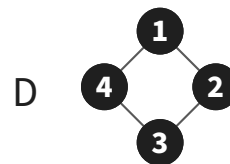
ON: _____

made / not yet



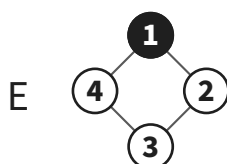
ON: _____

made / not yet



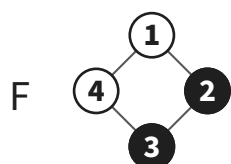
ON: _____

made / not yet



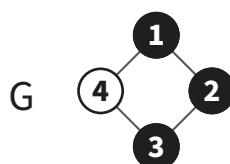
ON: _____

made / not yet



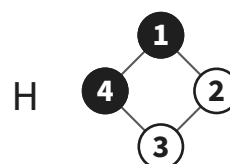
ON: _____

made / not yet



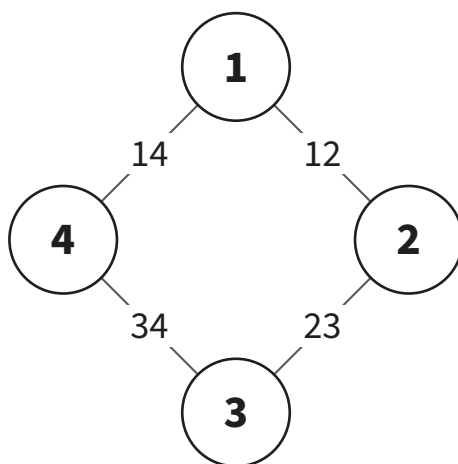
ON: _____

made / not yet



ON: _____

made / not yet

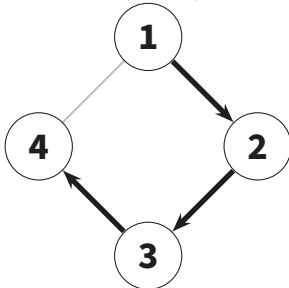


Compare the ON counts in the pictures you made.

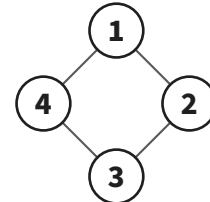
Point to what is alike. Then compare the “not yet” pictures.

Problem 20: Follow each arrow path on the big board: change both ends of each line, in order. Start all OFF for each path. Darken the ON lamps in its final picture.

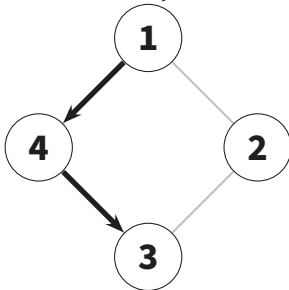
A: 12, 23, 34



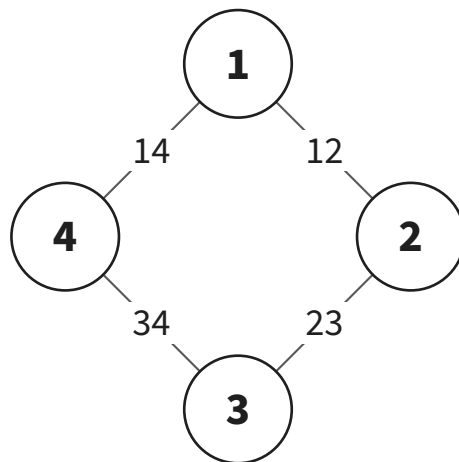
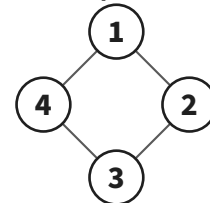
Final picture:



B: 14, 43

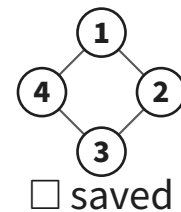
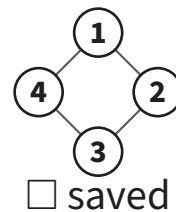
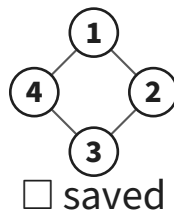
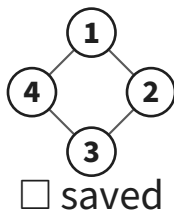
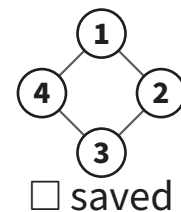
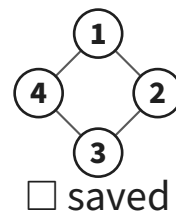
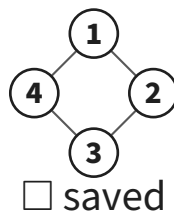
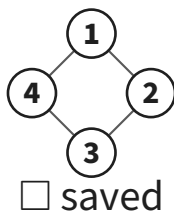
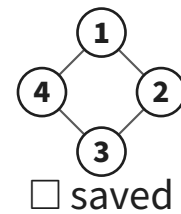
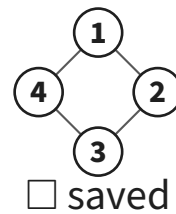
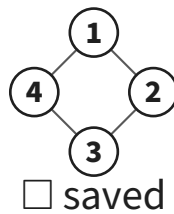
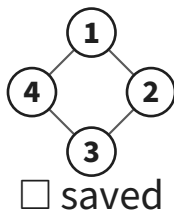
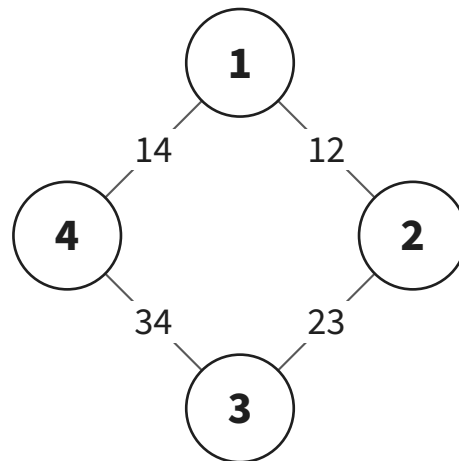


Final picture:



Replay one path slowly. Point to each lamp that changes twice. Point to each lamp changed just once.

Problem 21: Start all OFF on the big board. Try some moves. Copy the result onto a little ring: darken its ON lamps. Reset and make different pictures. Include all OFF. Leave extra rings empty.



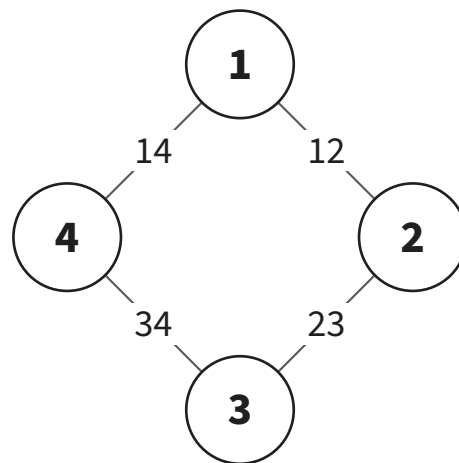
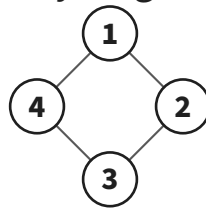
Check “saved” under each picture you record, including all OFF. This keeps unused rings separate.

Problem 22: Look at the pictures you saved in Problem 21. Count the ON lamps in each. Write that count beside its little ring. Point to the pictures with the same count. Count how many different pictures you have.

I saved _____ different pictures.

Choose a picture whose moves you have not written down. Copy it here. Start all OFF on the big board and find a move list that makes it.

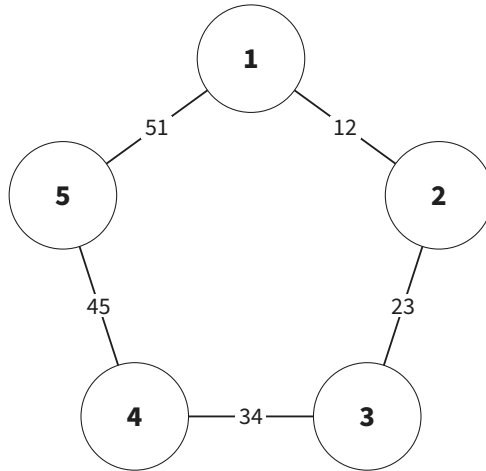
My target:



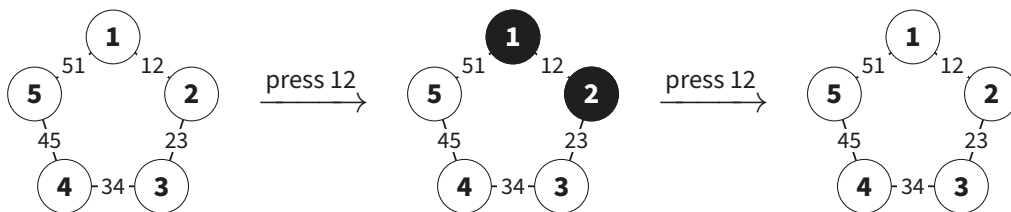
My moves: _____

Reset all OFF and ask a partner to replay your list. Compare the result with your target.

Problem 23: Try a five-lamp ring. Start all OFF on the large board. Press a line: change both end lamps. Add or erase pencil dots, or flip two-sided counters. OFF becomes ON; ON becomes OFF.



Write **12** for the line joining lamps 1 and 2. Dotted or dark means ON; white means OFF. You may ask someone to write your moves while you point.



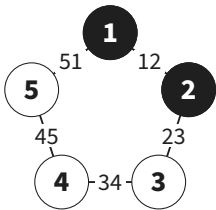
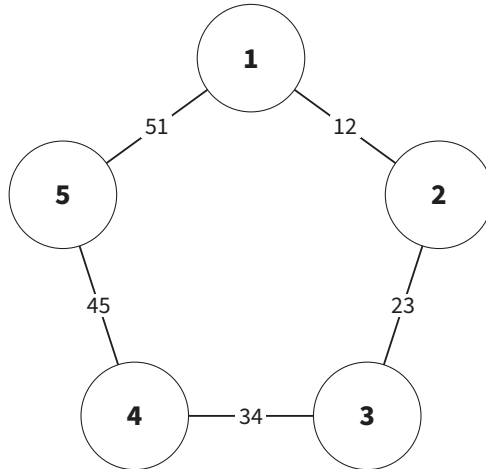
Try the two presses shown above. Then start all OFF and choose two presses of your own. Write them in the order you press them.

First press: _____

Second press: _____

Get back to all OFF. Write the presses you used.

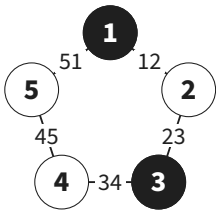
Problem 24: Start all OFF for each target below. Make the dark lamps ON and the white lamps OFF. Write two different move lists for each target. Circle the shorter list, or write “same” if they tie.



Make this picture.

First list: _____

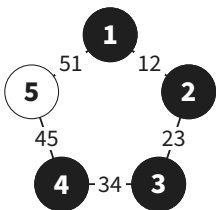
Another list: _____



Make this picture.

First list: _____

Another list: _____



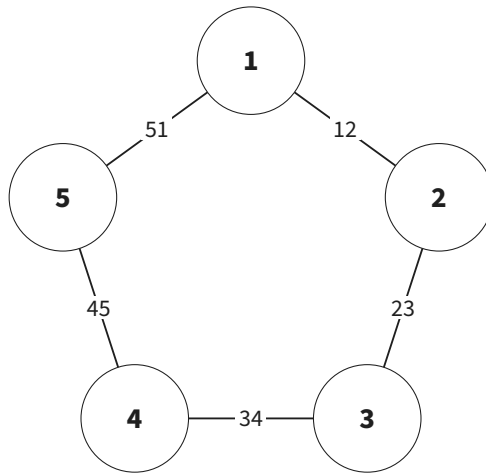
Make this picture.

First list: _____

Another list: _____

Choose one target. Test single presses, resetting to all OFF before each press. Can one press make that target? Tell a partner what happened.

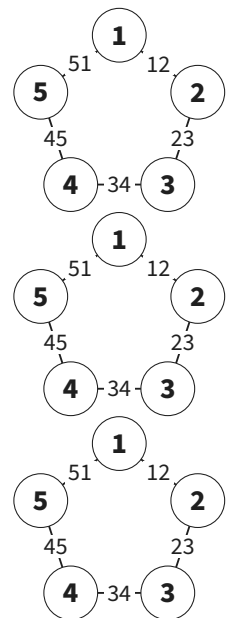
Problem 25: Start all OFF for each move list. Read from left to right. Press one line at a time, then color the lamps that finish ON in its small ring.



Press **12**, then **23**.

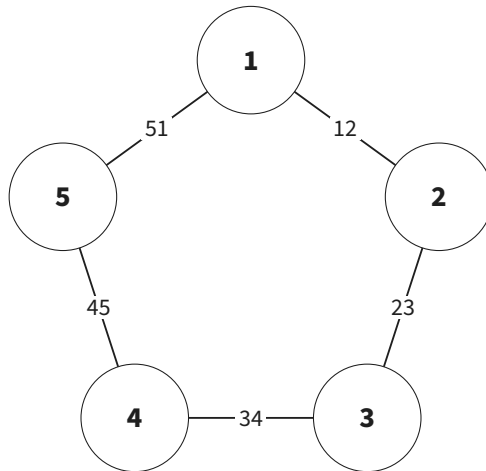
Press **23**, then **12**.

Press **12**, then **23**, then **12**, then **12**.



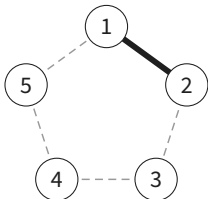
Compare the three pictures. In the last list, cross out two copies of the same press. Reset and try the presses left over. Do you get the same picture?

Problem 26: For each pair, start all OFF. Press each thick line on the left once, then color the lamps that finish ON. Reset to all OFF. Test the thick lines on the right and color that final picture.



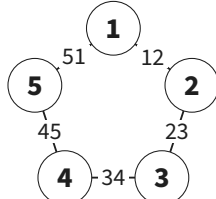
The dashed lines are the lines you do not press. In the first pair, try line 12 on the left; on the right try 23, 34, 45, 51.

Press these

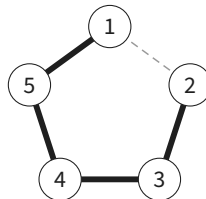


___ presses

Color result

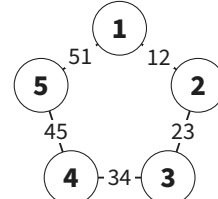


Reset. Press these

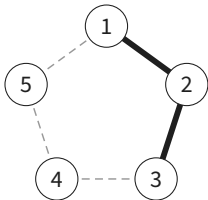


___ presses

Color result

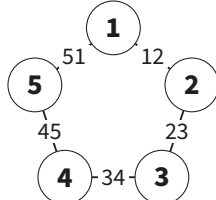


Press these

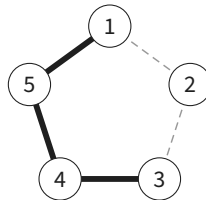


___ presses

Color result

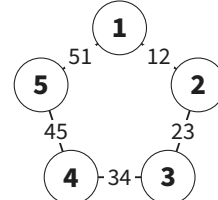


Reset. Press these



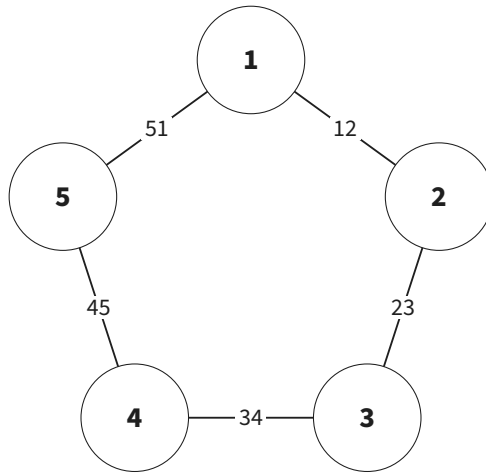
___ presses

Color result

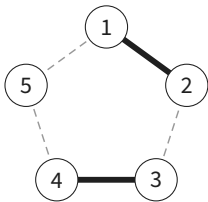


In each pair, count the presses and compare the two final pictures. Circle the side with fewer presses.

Problem 26 (continued): Try two more pairs. Begin all OFF each time. Press each thick line once and color the lamps that finish ON. Count the presses.

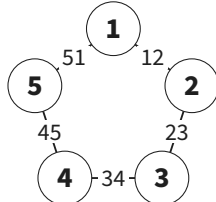


Press these

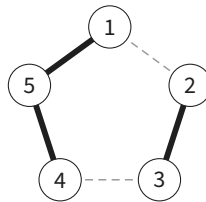


___ presses

Color result

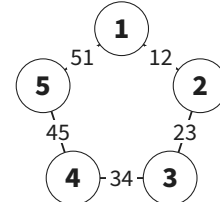


Reset. Press these

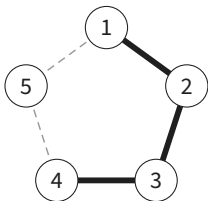


___ presses

Color result

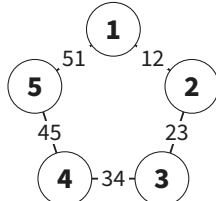


Press these

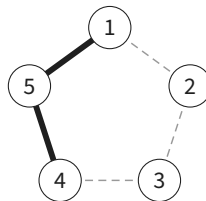


___ presses

Color result

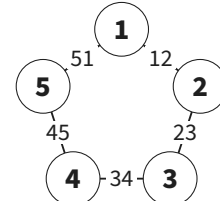


Reset. Press these



___ presses

Color result

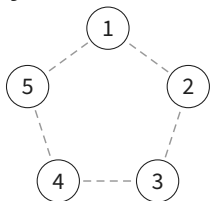


Compare the two final pictures in each pair. Circle the side with fewer presses.

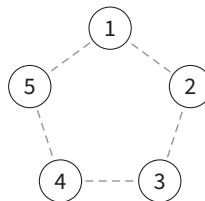
Finish this sentence with words or a drawing: “When I press the other lines, ...”

Problem 27: Make your own pair of line pictures. Choose one, two, or three lines on the left ring and trace them thickly. On the right ring, trace every line you did not choose.

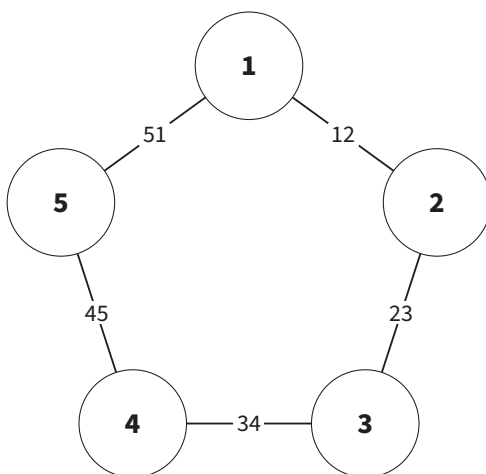
My chosen lines



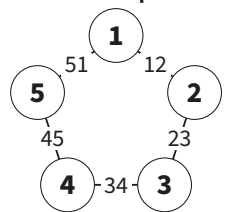
The other lines



Start all OFF on the large mat. Press each chosen line once. Color the first final picture below. Reset and test the other lines; color the second final picture.

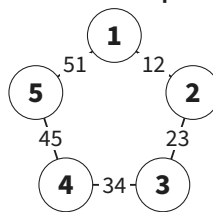


First final picture



_____ presses

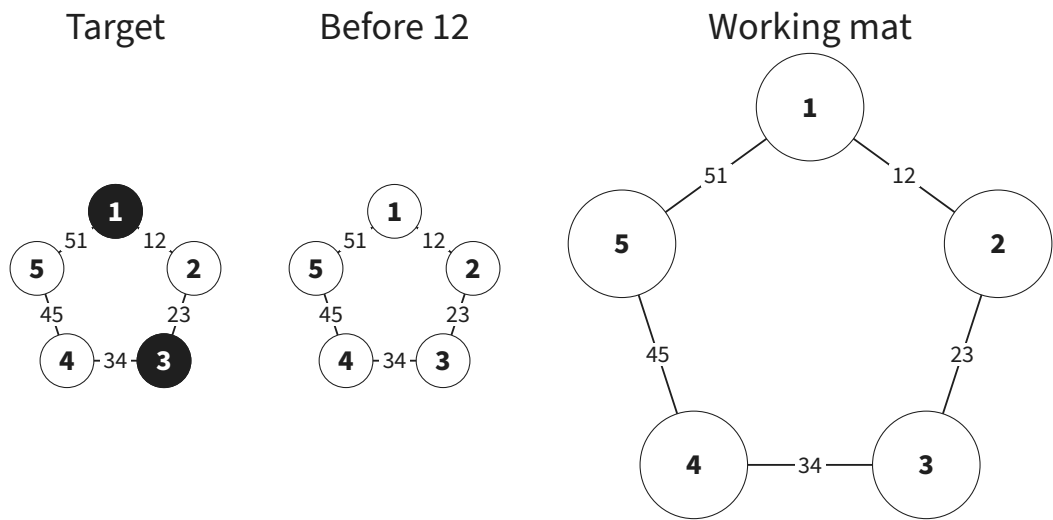
Second final picture



_____ presses

Compare your pictures. Show your partner the pair and ask them to test it.

Problem 28: Make the target below by fixing one lamp at a time. Start all OFF. This time, do not press line 51. Use each line at most once.



Lamp 1 is OFF, but the target needs it ON. **Press 12 now.** Circle “yes” in the first row. Lamp 1 now matches the target.

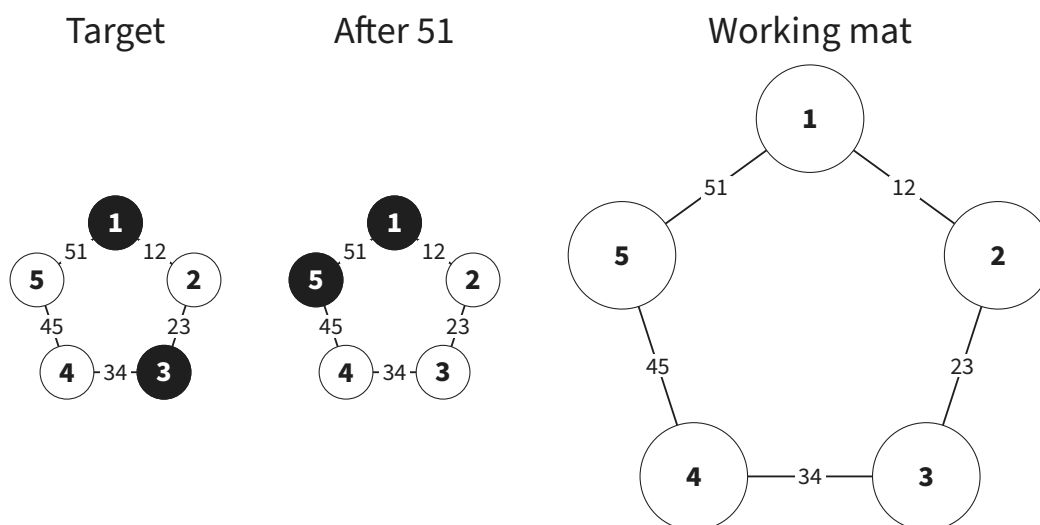
Next look at lamp 2. If it differs from the target, press 23; if it matches, leave 23 alone. Do the same for lamps 3 and 4, using the next line each time.

Look at lamp	To match the target, press this line?	Circle one	
1	12	yes	no
2	23	yes	no
3	34	yes	no
4	45	yes	no

Stop after 45. Does lamp 5 match the target? ☐ yes ☐ no

Write the lines you pressed, in order.

Problem 28 (continued): Try the same target again. Start all OFF. This time, press line 51 first. Then decide about 12, 23, 34, 45, in that order. Use each line at most once.



After pressing 51, lamp 1 is already ON, just as the target needs. **Do not press 12.** Circle “no” in the first row.

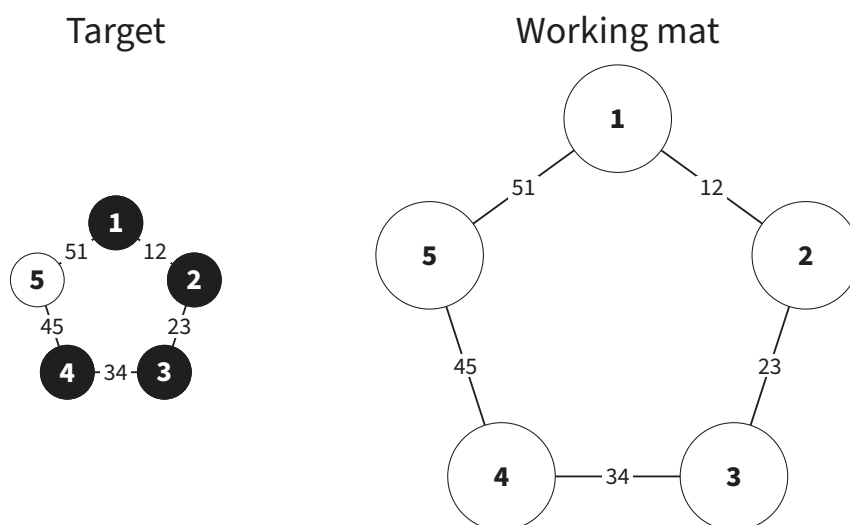
For each next lamp, compare it to the target. Press the next line only if that lamp needs to change.

Look at lamp	To match the target, press this line?	Circle one	
1	12	yes	no
2	23	yes	no
3	34	yes	no
4	45	yes	no

Stop after 45. Does lamp 5 match the target? ☐ yes ☐ no

Write your whole move list, starting with 51.

Problem 29: Make this new target in two ways. Start all OFF for each trial. Decide about 51 first, then 12, 23, 34, 45. Press each line at most once.



Look at each lamp in order. If it differs from the target, press the line below it. If it matches, do not press that line. After lamp 4, stop and check lamp 5. Include 51 in your press count if you used it.

First trial: leave 51 alone.

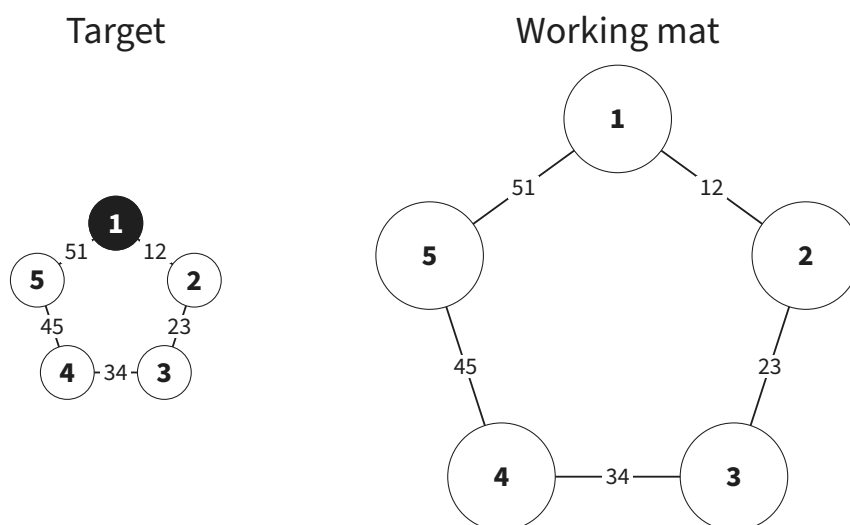
Lamp to fix	1	2	3	4
Line to press?	12	23	34	45
Circle one	yes / no	yes / no	yes / no	yes / no
Lamp 5 matches the target: ☐ yes ☐ no				Total presses: _____

Reset to all OFF. Second trial: press 51 first.

Lamp to fix	1	2	3	4
Line to press?	12	23	34	45
Circle one	yes / no	yes / no	yes / no	yes / no
Lamp 5 matches the target: ☐ yes ☐ no				Total presses: _____

Circle the trial that uses fewer presses. Point to its lines on the working mat.

Problem 30: Try this target: only lamp 1 ON. Start all OFF for each trial. Choose about 51 first, then decide about 12, 23, 34, 45. Press each line at most once.



At each lamp, press the next line only if that lamp needs to change to match the target. After deciding about 45, stop. Check lamp 5 carefully. Include 51 in your press count if you used it.

First trial: leave 51 alone.

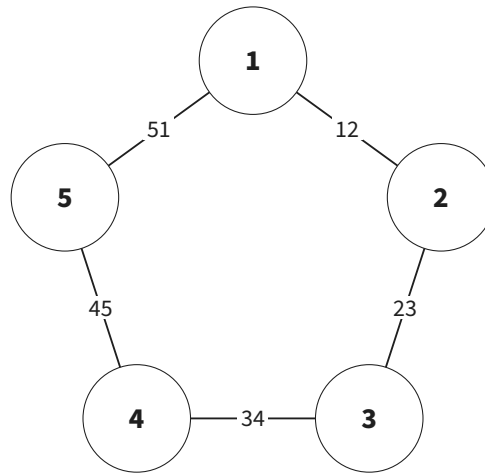
Lamp to fix	1	2	3	4
Line to press?	12	23	34	45
Circle one	yes / no	yes / no	yes / no	yes / no
Lamp 5 matches the target: ☐ yes ☐ no				Total presses: _____

Reset to all OFF. Second trial: press 51 first.

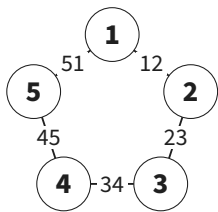
Lamp to fix	1	2	3	4
Line to press?	12	23	34	45
Circle one	yes / no	yes / no	yes / no	yes / no
Lamp 5 matches the target: ☐ yes ☐ no				Total presses: _____

Put a cross beside a trial if its final picture does not match the target. Show a partner which lamp does not match.

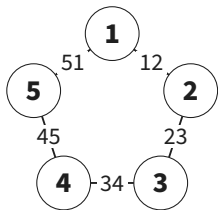
Problem 31: Invent three new targets. In each small ring, color two or four lamps ON. Test each target on the large mat, starting all OFF each time.



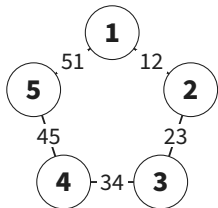
Find one move list using each line at most once. For the other list, use every line missing from the first list, once each. Test both. Count the presses and circle the shorter list.



My target. Counts: _____
 First list: _____
 Another list: _____



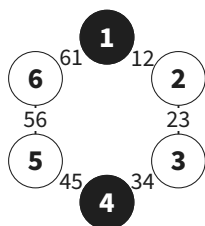
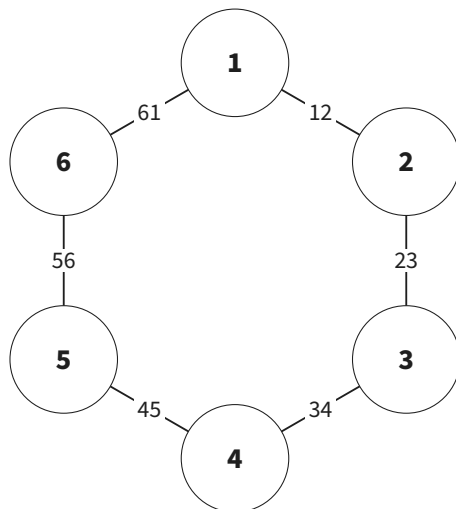
My target. Counts: _____
 First list: _____
 Another list: _____



My target. Counts: _____
 First list: _____
 Another list: _____

If you want another puzzle: can you draw a target with two or four ON lamps that needs more than two presses?

Problem 32: Start all OFF on this six-lamp ring. Follow one route between the dark lamps of a target. Press each line on that route once. Reset and try the other route. Make both targets below.

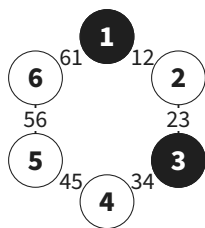


First route: _____

Press count: _____

Other route: _____

Press count: _____



First route: _____

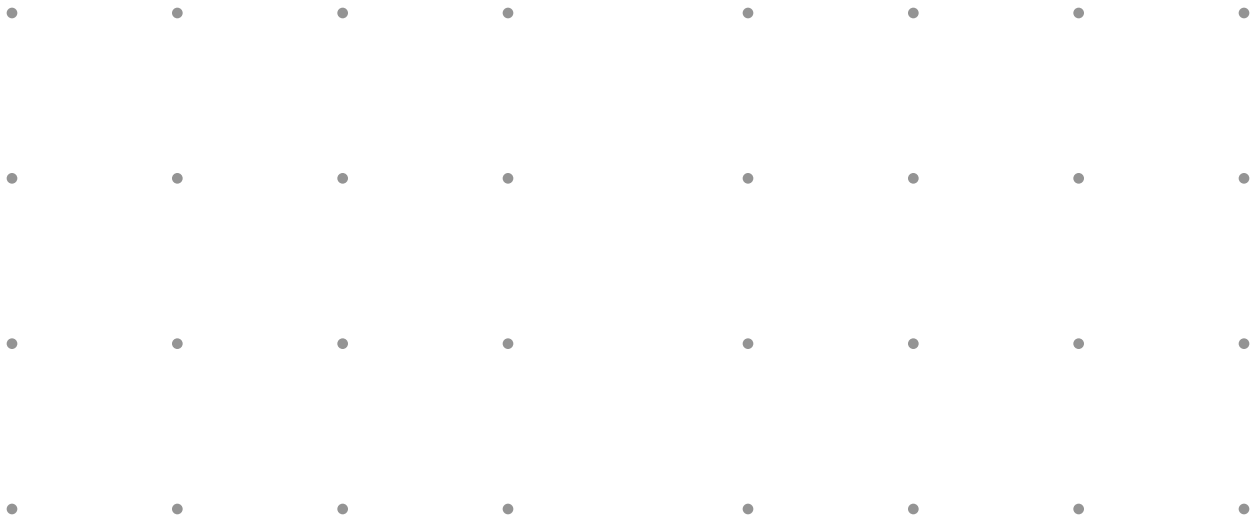
Press count: _____

Other route: _____

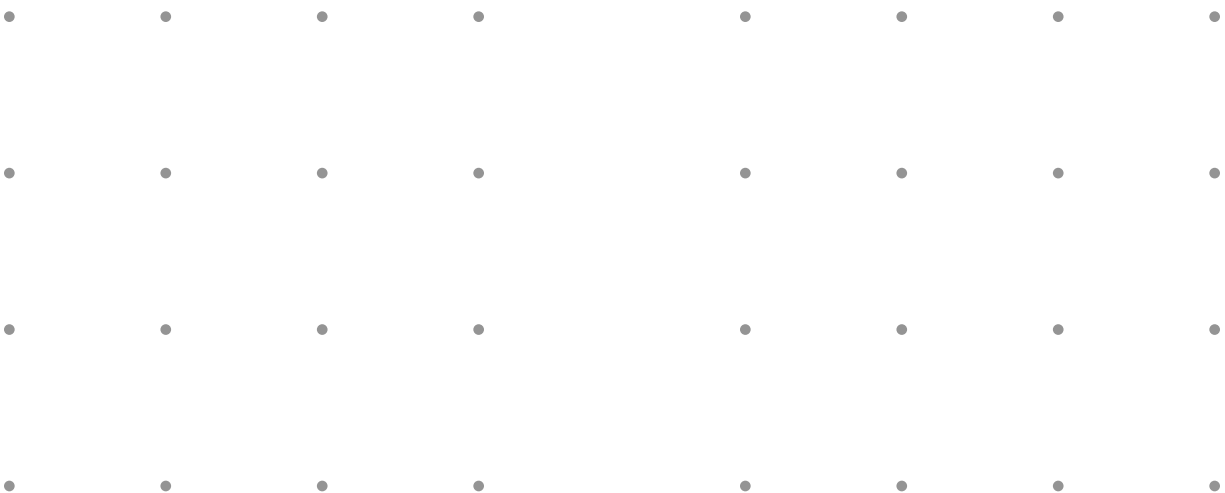
Press count: _____

Circle each pair of routes that uses the same number of presses. Could two routes around the five-lamp ring use the same number? Draw or write your idea.

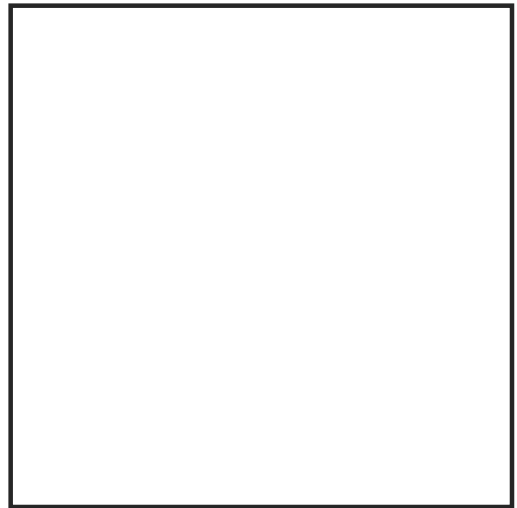
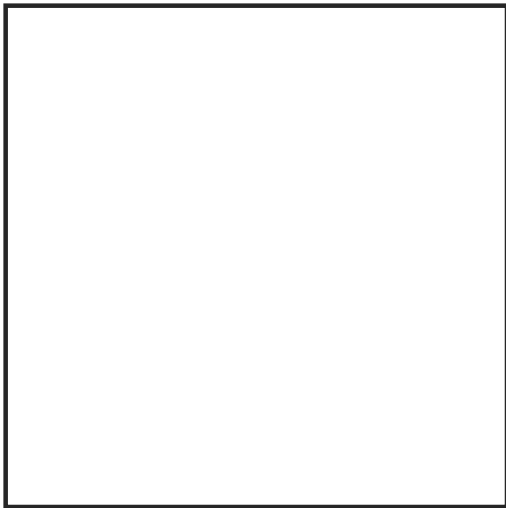
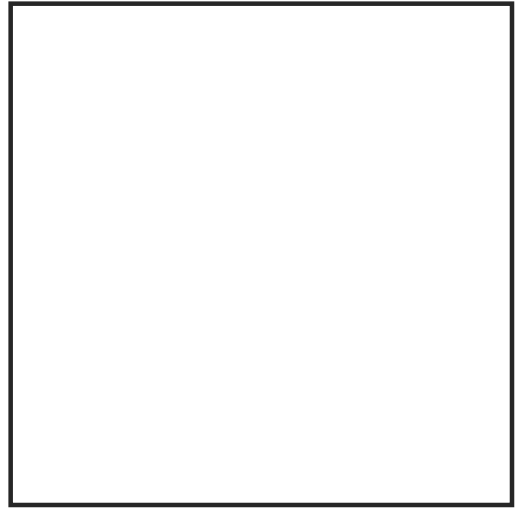
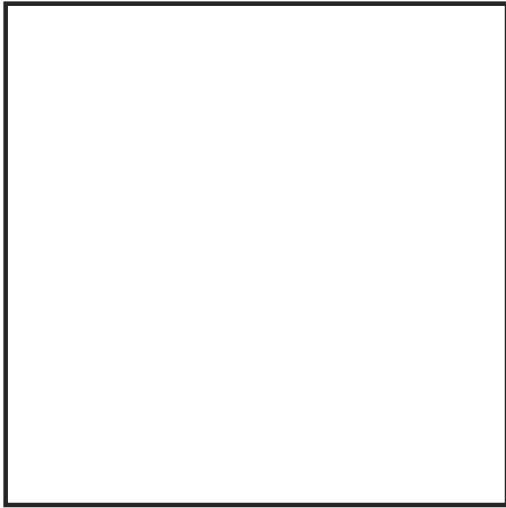
Problem 33: Join dots with straight lines to make a closed shape with THREE corners. Keep your lines from crossing. Make two different shapes.



Problem 34: Now make two different closed shapes with FOUR corners. Touch each corner where your line turns.

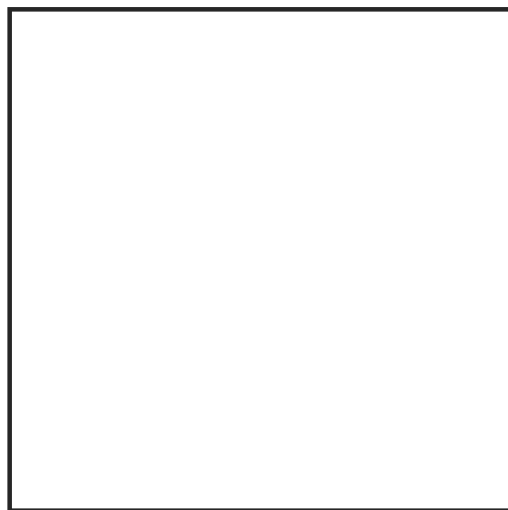
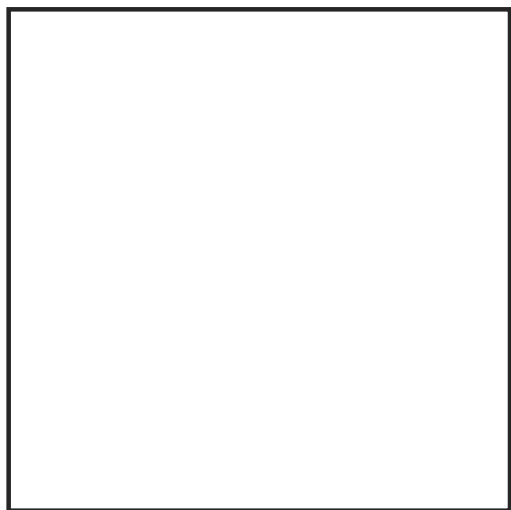
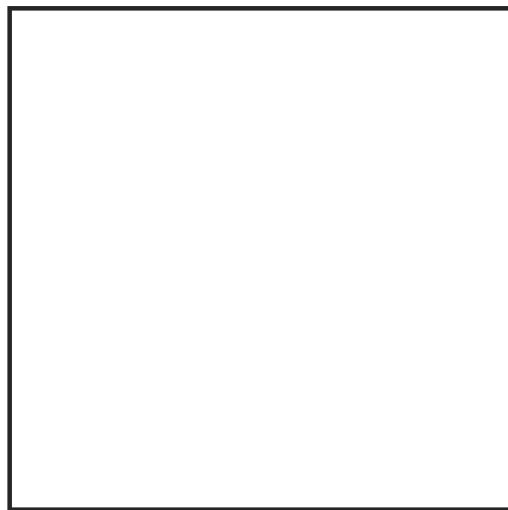
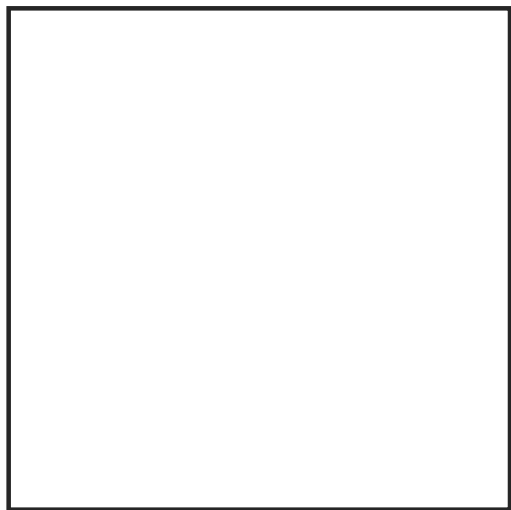


Problem 35: Each square is a house. Draw ONE straight wall all the way across each house, from its outside edge to its outside edge. Make two rooms. Try different walls.

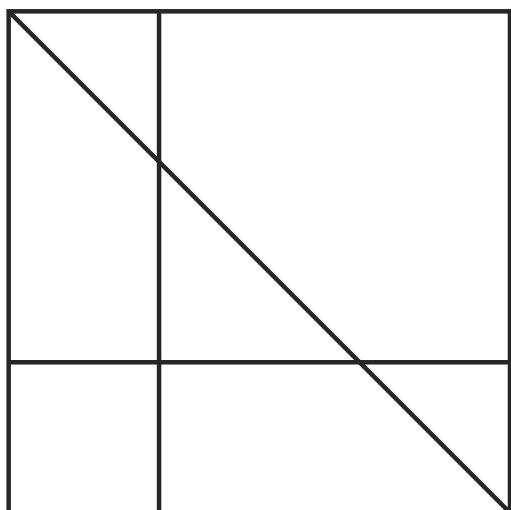
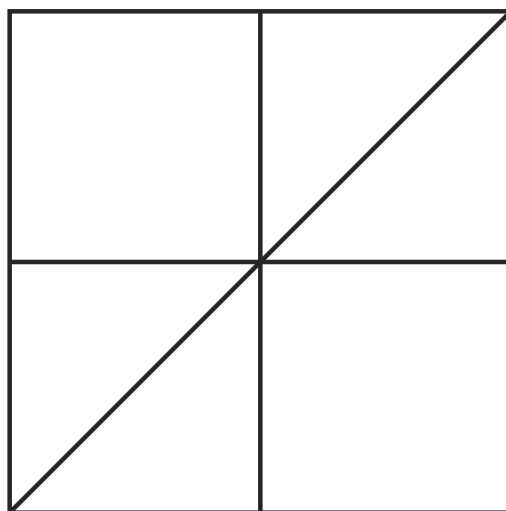
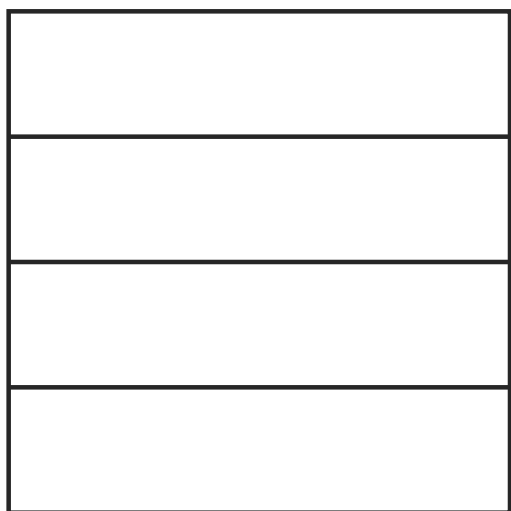


Problem 36: Make or find a house with two triangle rooms. Make or find one with two four-corner rooms. Draw on another sheet if you want to try more houses.

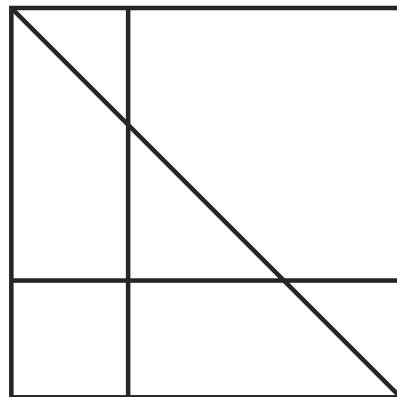
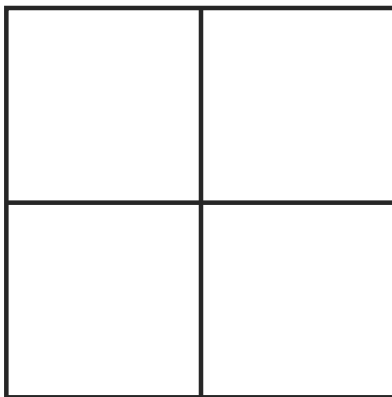
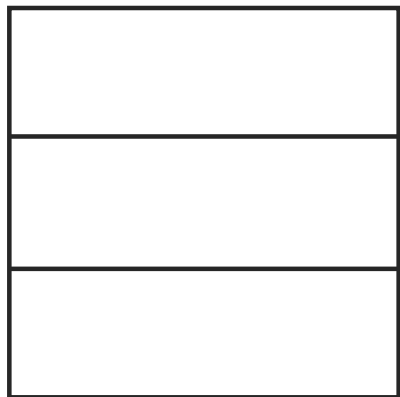
Problem 37: Draw TWO different straight walls all the way across each house. Walls may cross. Put one dot in each room. Try to make different numbers of rooms.



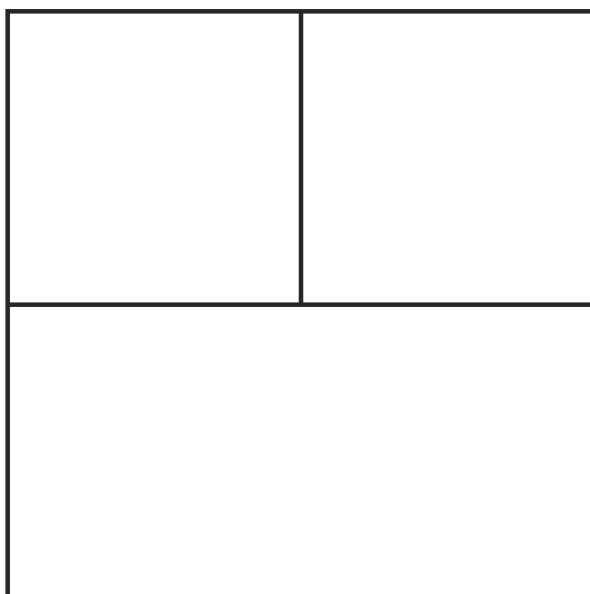
Problem 38: These houses each have THREE straight walls. Put one dot in every room. Which has the most rooms? In the empty house, try three different walls of your own.



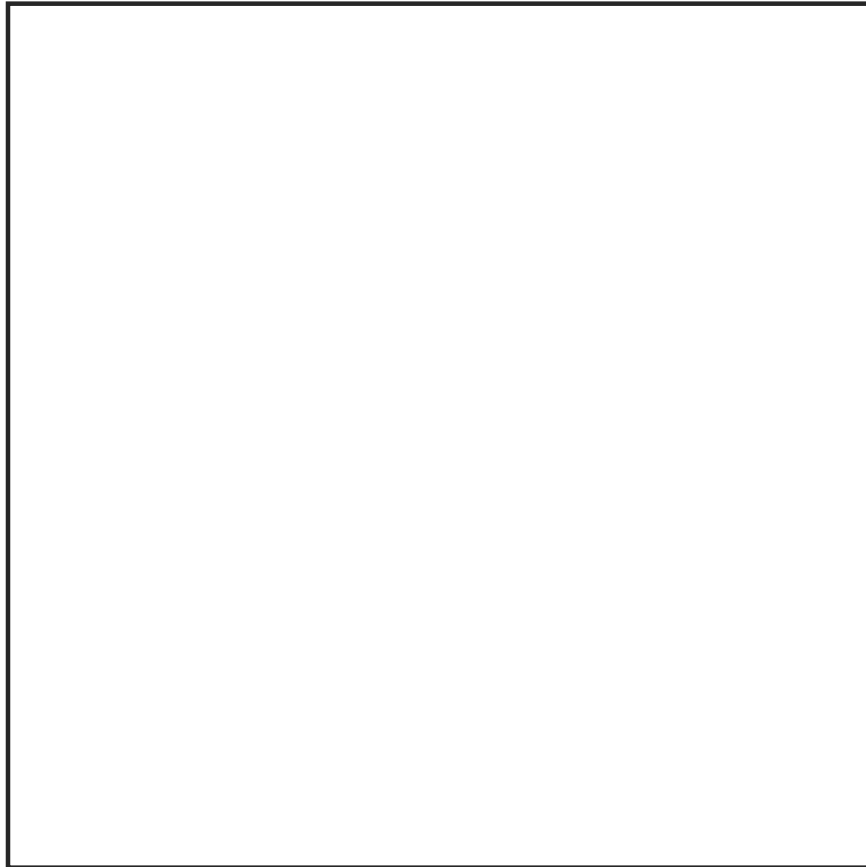
Problem 39: Put a dot or an X in every room in the first three houses. Rooms sharing a wall must have DIFFERENT marks. Rooms touching only at a corner may match.



Problem 40: Try this house. If two marks do not work, you may also use an O.

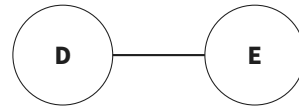
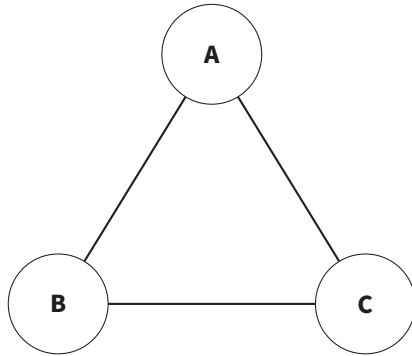


Problem 41: Make a house with one, two, or three different straight walls. Draw each wall all the way across. Let a partner put one dot in every room. Swap jobs.

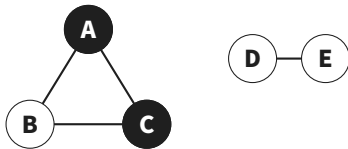


Problem 42: Make a giant house together on a whiteboard or another sheet. Take turns adding one new straight wall all the way across. Do not trace an old wall. Stop after three walls. Find every room together.

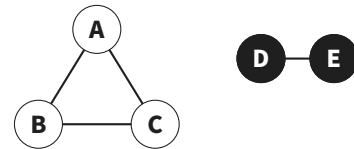
Problem 43: Start all OFF on the large mat. Press a line to flip the lamps at both ends. Make each target below: dark means ON; white means OFF. Use dots you can erase or counters you can flip.



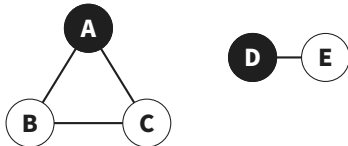
Write **AC** for a press of the line joining A and C. A move cannot jump across the gap. Reset to all OFF for every target.



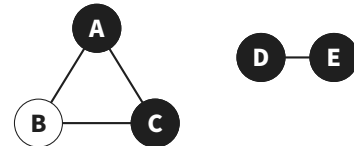
Moves, or “not yet”:



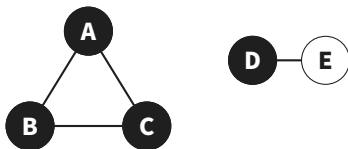
Moves, or “not yet”:



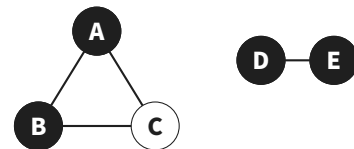
Moves, or “not yet”:



Moves, or “not yet”:

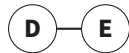
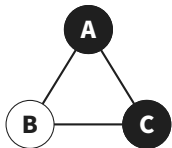
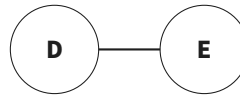
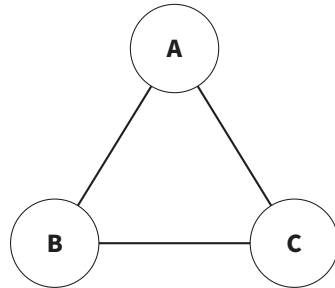


Moves, or “not yet”:

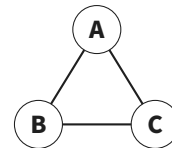


Moves, or “not yet”:

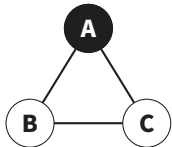
Problem 44: Try these targets, starting all OFF each time. In each picture, count the dark lamps in the triangle. Then count the dark lamps in the pair. Write the two counts and circle “made” or “not yet.”



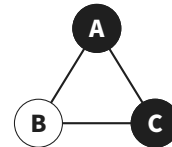
ON in triangle: ____ in pair: ____
Made / not yet



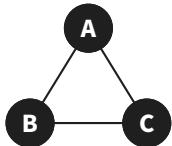
ON in triangle: ____ in pair: ____
Made / not yet



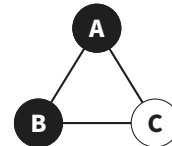
ON in triangle: ____ in pair: ____
Made / not yet



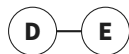
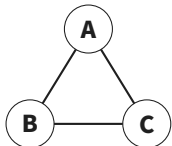
ON in triangle: ____ in pair: ____
Made / not yet



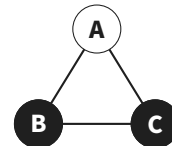
ON in triangle: ____ in pair: ____
Made / not yet



ON in triangle: ____ in pair: ____
Made / not yet



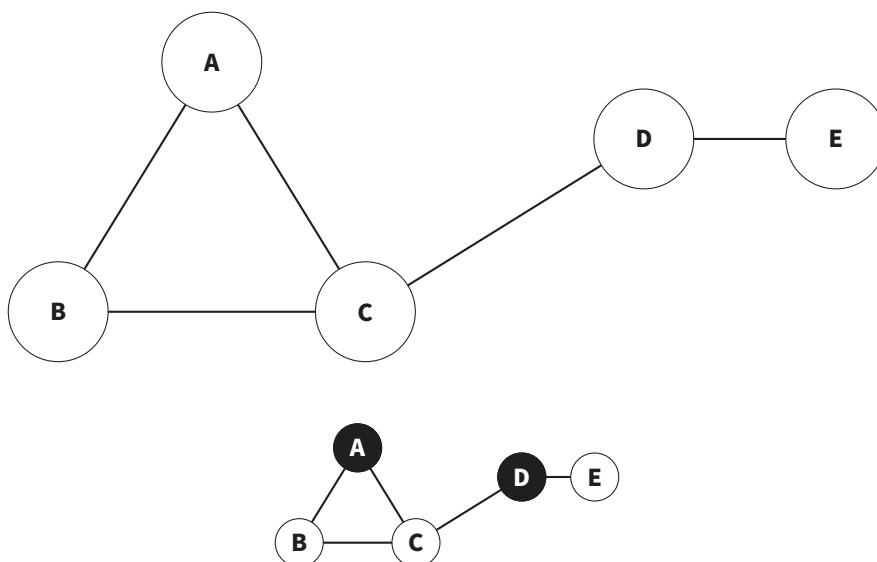
ON in triangle: ____ in pair: ____
Made / not yet



ON in triangle: ____ in pair: ____
Made / not yet

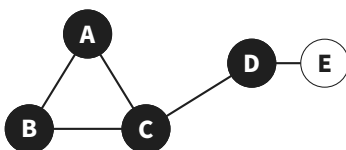
Point to the pictures you made. What do you notice about their two counts? Show a partner a rule that predicts which pictures can be made.

Problem 45: The two pieces now have a bridge: line CD. Start all OFF and try each target below. Press a line to flip both ends. Write the lines you press in order.



Make this picture. Moves: _____

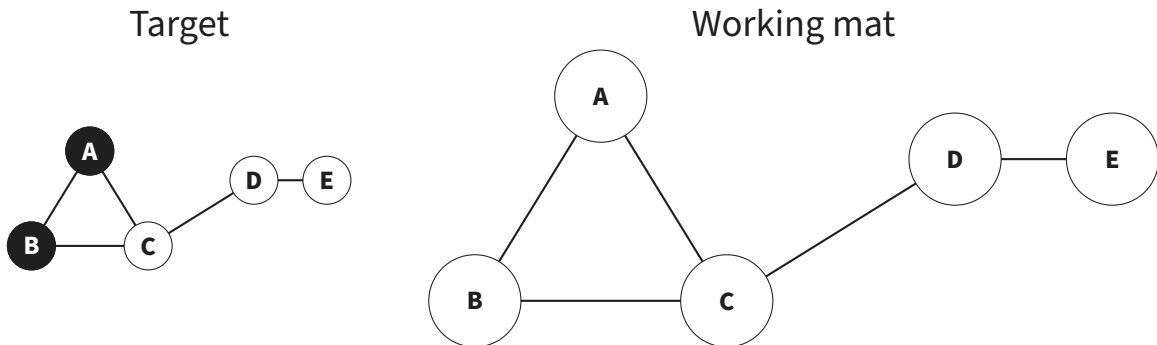
Reset to all OFF. Make the next picture.



Moves: _____

Point to a line your solution needs that was missing from Problem 43. What changed when we added it? Tell a partner, or draw your idea.

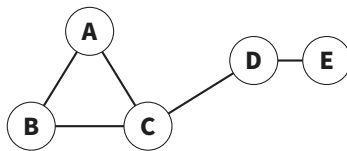
Problem 46: Make the pictured target by trying the two routes shown below. Begin all OFF on the large mat. Press the lines in the first route, then the lines in the second route.



First route: A–C–D. Then: B–C–D.

Try the full list: **AC, CD, BC, CD.**

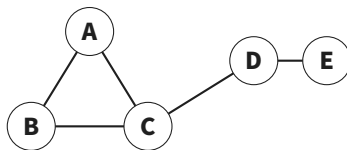
Color the lamps that finish ON in this small picture.



In the list below, cross out both copies of the repeated press.

AC CD BC CD

Reset to all OFF. Try only the presses you did not cross out. Color that final picture below.



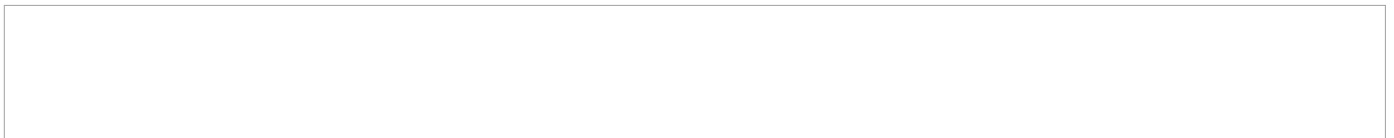
Compare your two pictures with the target. Do both lists work?

Find a longer list of your own that makes this target. Give it to a partner to shorten and test.

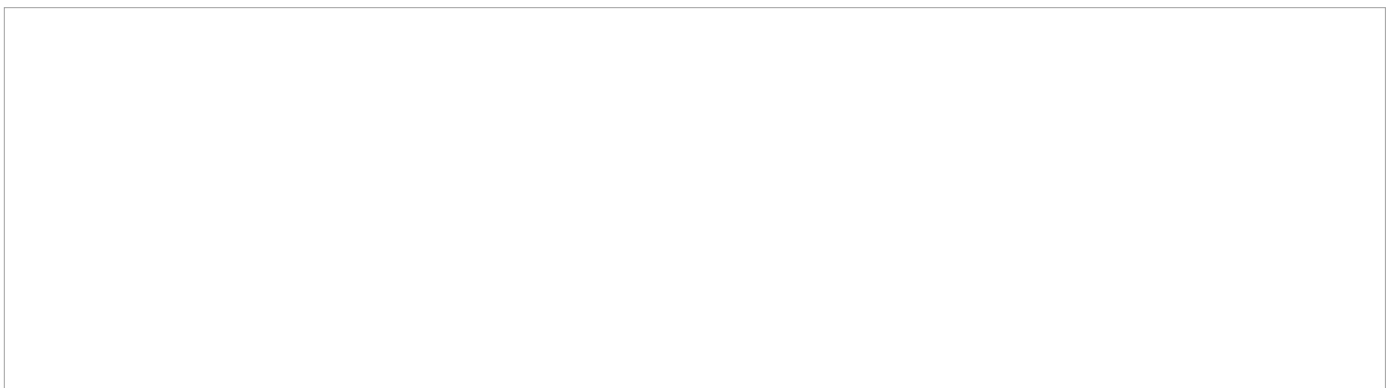
Problem 47: Invent a network with two separate pieces. Draw at least three lamps in each piece, joined by lines. Leave a gap with no line between the pieces.



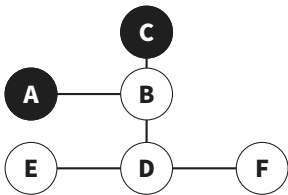
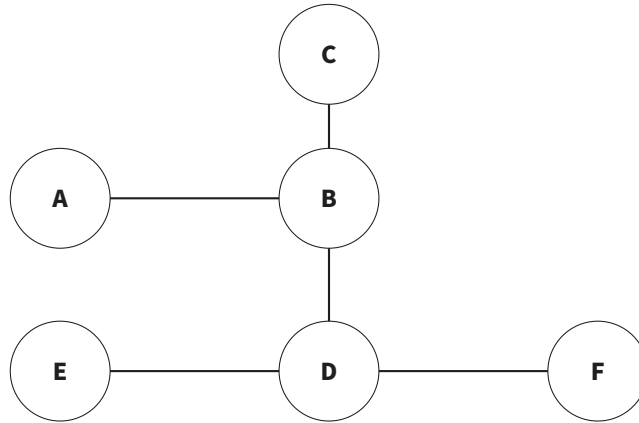
Choose one target you predict can be made from all OFF. Show it by coloring lamps in your drawing. Try it with dots on a whiteboard, or counters on this page. Write a move list or trace the lines you used.



Copy your network below. Color a target you predict cannot be made from all OFF. Ask a partner to try it. Point to the lamps that helped you choose it.



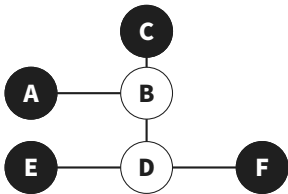
Problem 48: This network has no closed loop. Start all OFF. Press a line to flip both ends. Make each pictured target using every line at most once. Use dots you can erase or counters you can flip.



Make this picture.

Moves: _____

Reset to all OFF before making the next picture.

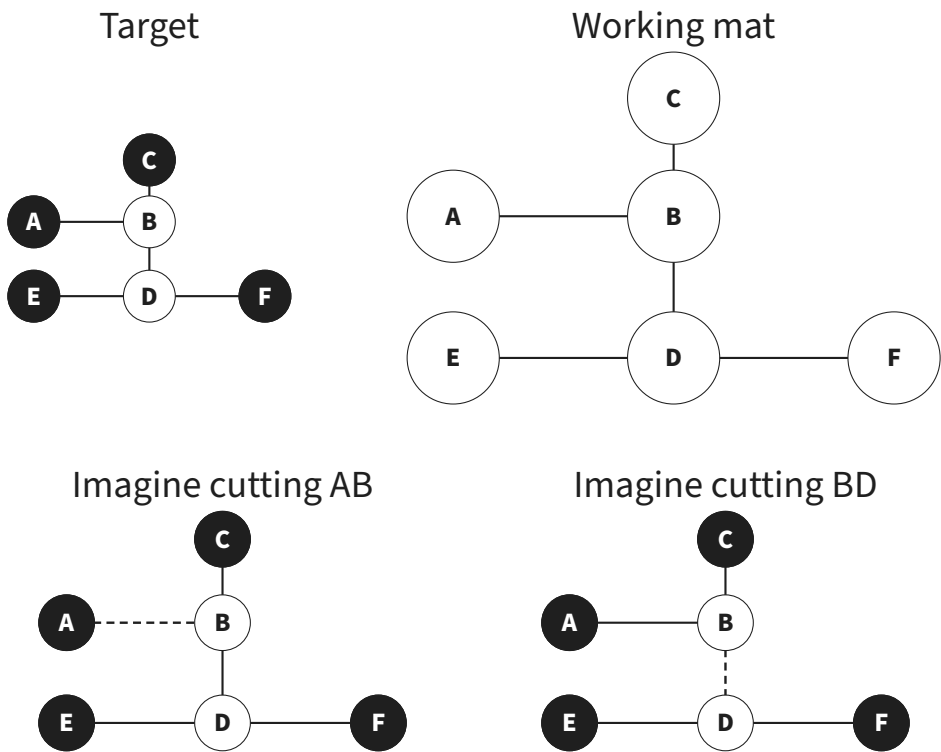


Make this picture.

Moves: _____

Choose one target. Trade move lists with a partner and replay their list. Point to any lines you both used.

Problem 49: Use the target shown below. Imagine cutting one line at a time. Count the dark target lamps on the named side of the cut. Predict which lines to press, then test from all OFF.



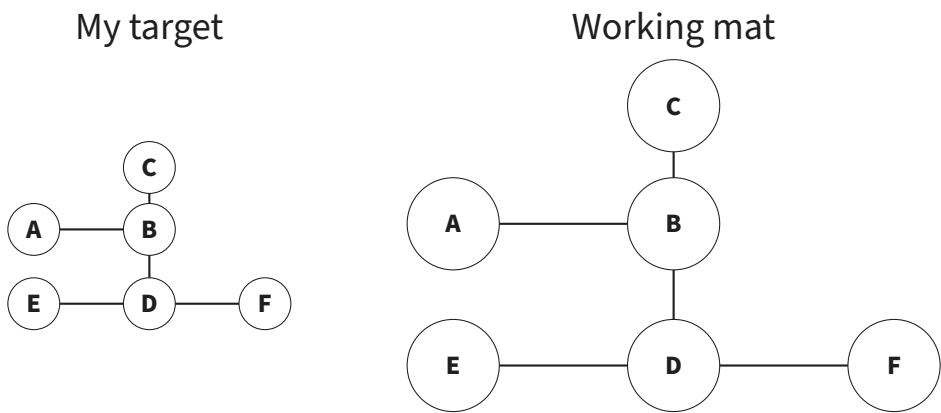
At cut AB, the A side has **1** target lamp. Press AB once. At cut BD, the A,B,C side has **2** target lamps. Leave BD alone.

Complete the other rows. Use each line at most once.

Cut line	Count on this side	Target count	Press?
AB	A	1	yes
BC	C	—	yes / no
BD	A, B, C	2	no
DE	E	—	yes / no
DF	F	—	yes / no

Try your predicted list. Does it make the target? Show a partner how the counts helped you choose the lines.

Problem 50: Invent another target on this tree. Color 2, 4, or 6 lamps ON in the small picture. Choose a different target from Problem 48. Leave the large working mat all OFF.



Before trying moves, imagine cutting each line. Count the ON lamps in your target on the named side. Predict whether to press the line. Use each line at most once.

Cut line	Count on this side	Target count	Press?
AB	A	_____	yes / no
BC	C	_____	yes / no
BD	A, B, C	_____	yes / no
DE	E	_____	yes / no
DF	F	_____	yes / no

Write your predicted move list: _____

Now test it from all OFF. Compare the result with your target. If they differ, circle a row to check again.

Trade targets with a partner. Predict the presses for their target and test them together.

Problem 1: Use five letter cards. One turn changes every letter using the chosen key, keeping message positions fixed. Start each experiment with ABCDE. Repeat the same key and record each message until the whole row first returns. Circle that turn.

Input	A	B	C	D	E
Key R	B	C	D	E	A
Key S	B	C	A	E	D

Turn	R message	S message
0	ABCDE	ABCDE
1		
2		
3		
4		
5		
6		

Problem 2: For each key, follow A and draw its arrows until it comes back. Then start at a letter not yet drawn. Draw every loop, including any letter that stays put. Compare the two keys' loop sizes and return times.

Problem 3: Build three more keys from five letter cards: loops of sizes 4+1, then 2+2+1, then 3+1+1. Draw each key and test when all letters first return.

Loop sizes	My arrows	Return
4 + 1		
2 + 2 + 1		
3 + 1 + 1		

Problem 4: Return to key S, with loops of sizes 3 and 2. Move a counter around each loop, one arrow per turn. In each row below, circle every turn when that counter is home. Box the first positive turn circled in both rows.

Loop	1	2	3	4	5	6	7	8	9	10	11	12
Size 3	1	2	3	4	5	6	7	8	9	10	11	12
Size 2	1	2	3	4	5	6	7	8	9	10	11	12

Problem 5: Invent another five-letter key. Predict its return from its loops, then test. Record your prediction and result.

Problem 6: Arrange five letter cards into loops. Organize different arrangements by their largest loop: 5, then 4, then 3, then 2, then 1. For each largest loop, split the remaining cards in every way you can. Record sizes and the first common return. Changing letter names does not create a new list of sizes.

Loop sizes	First common return

Problem 7: Choose the arrangement with the longest return. Write a key that has those loops. Test it with counters and mark its return turns. Compare with your earlier keys.

Problem 8: Key P swaps A and B. Key Q swaps B and C. Both leave all other letters alone. Start with ABCDE each time. Apply the first key, record the whole row, then apply the second key to that row.

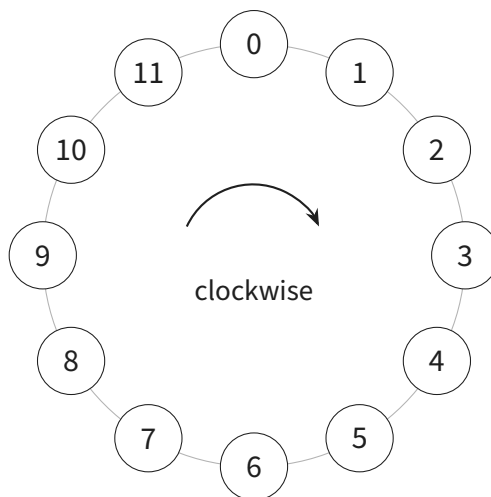
Order	After first key	After second key
P then Q		
Q then P		

Problem 9: Treat P then Q as one turn. Repeat that whole turn on ABCDE until the row first returns. Draw the loops of this combined key.

Problem 10: Change Q to a new key T that swaps D and E and fixes A,B,C. Test P then T and T then P. Compare this pair with Problem 8.

Problem 11: Start with ABCDE and apply P then Q. Now find an order of these same two keys that undoes the result. Record the full check.

Problem 1: A shift $+k$ moves k places clockwise, wrapping after 11. Try $+4$ from 0, then from 9. Record both landings. Find where inputs 2 and 8 go under the same shift.



Problem 2: A secret shift takes 9 to 1. Use the ring to recover it. Test the additional clue 2 to 6. Decode outputs 1,4,10 by undoing that shift.

Problem 3: Repeat $+4$ from 0 until it first returns. Write the whole route. Start at a position not yet used and repeat. Continue until every position is in a recorded loop.

Problem 4: Use the twelve-place ring. Start at 0 and repeat +8 until you return. Find its other loops. Compare the loop members and their order with +4.

Problem 5: Try +3, then +6, then +5. For each rule, find the route from 0 and all remaining loops. Record the first return time and the number of loops.

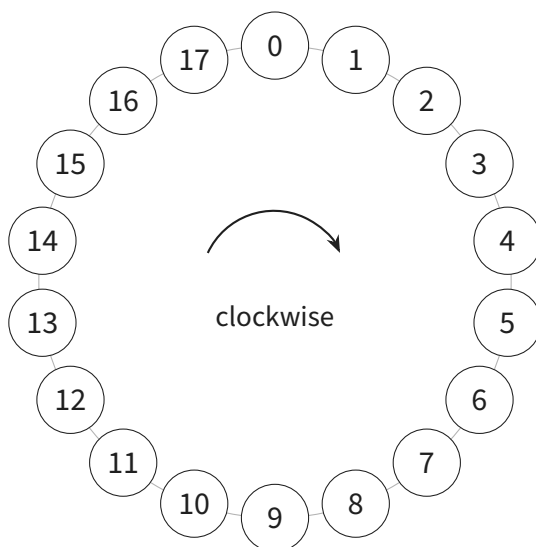
Shift	Routes	Return	Count
+3			
+6			
+5			

Problem 6: Compare these shifts with +4 and +8. Group shifts with the same return time. Record a prediction connecting the step size, ring size, and number of loops.

Problem 7: Test the remaining shifts on the twelve-place ring. Record first positive return times and number of loops. Use your earlier results where you have them. For +0, a turn leaves the counter in place.

Shift	Return	Loops	Shift	Return	Loops
0			6		
1			7		
2			8		
3			9		
4			10		
5			11		

Problem 8: On this eighteen-place ring, predict the first return for +6 and +5. Test both and record their routes from 0. Compare with your prediction.



Problem 9: Each +4 turn on twelve places moves four spaces. Fill the distance table and circle the first positive total that is a whole number of laps. Repeat for +8. Compare this distance record with the routes you drew.

Turns	+4 distance	Whole laps?	+8 distance	Whole laps?
1				
2				
3				
4				

Problem 10: Find a shift that undoes +4 on twelve places. Test it from 0,1,5, and 11. Record the four out-and-back trips.

Problem 11: Start at 0, then 2, then 9. Compare +4 followed by +7 with +7 followed by +4. Record both final landings for each start. Find one shift that does the same job.

Problem 12: Choose another pair of shifts on twelve places. Test both orders from two starts. Record your combined shift and a rule suggested by the experiments.

Problem 1: Build these rows with heights 1, 2, 3, and 4. Look from both ends. Count visible towers; a taller tower hides any shorter tower behind it.

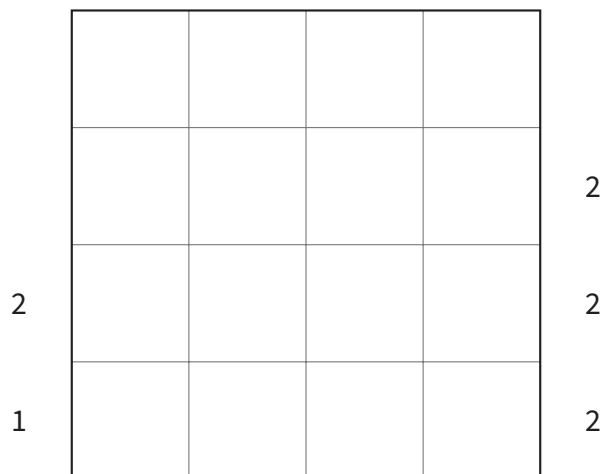
Order, left to right	Seen left	Seen right
1 4 2 3		
2 1 4 3		
4 1 2 3		
4 2 1 3		

Problem 2: Build a city using heights 1-4 once in every row and once in every column. Each outside number counts towers seen from that end. Solve the city below and record each tower height.

	4				
4					
					2
2					2
1					2

One tower I placed and the clue I used:

Problem 3: Remove the top clue 4 and the left clue 4. Rebuild using only the five remaining clues below. Keep a record of any different city you find.



Problem 4: Use four towers to test bottom rows before filling the city. Record every bottom row that shows one tower from the left and two from the right. Keep both possibilities available.

Possible bottom rows, left to right:

Problem 5: Now build rows showing two towers from each end. Record all the different rows you find. These are candidates for row 3, counting rows from the top.

Possible row 3 orders:

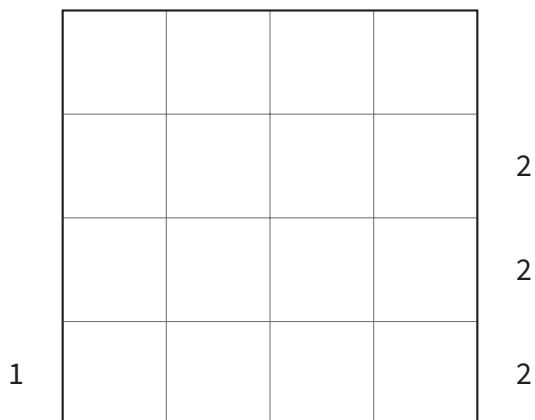
Problem 6: Pair each bottom row from Problem 4 with each row 3 from Problem 5. Reject a pair if a column repeats a height. Record the pairs that survive. For each pair, try every row 2 that avoids column repeats and shows two towers from the right. Write all the row 2 orders that work, or write “none.”

Bottom row	Row 3	Row 2 orders that work

Problem 7: For each case that survives Problem 6, fill the top row with the heights missing from its columns. Record the completed city. Check all five clues again.

				2
2				2
1				2

Problem 8: Start with the five-clue city from Problem 7. Cover the left clue on row 3. Try to make a different city that keeps the other four clues. Record it, then restore the covered clue.



Problem 9: Check the alternatives below one at a time. L3 means the left clue on row 3; R2 means the right clue on row 2. In each city, a slash begins the next row, from top to bottom. Check every row and column and the four retained clues. Record the actual view at the covered clue. Restore it before the next check.

Cover	Alternative city	Actual view
L3	1234 / 3412 / 2341 / 4123	
L4	1234 / 4123 / 3412 / 2341	
R2	1324 / 2431 / 3142 / 4213	
R3	1324 / 3142 / 2431 / 4213	
R4	1234 / 2143 / 3412 / 4321	

Problem 10: Remove every old clue. Use only left clues 4, 3, and 2 on rows 1, 2, and 3. Build a city with these clues and record its tower heights.

4				
3				
2				

Problem 11: List the top rows that work with its clue. Then list the row 2 orders that fit its clue and avoid repeats under the top row. Continue with row 3. Fill row 4 with each column's missing height.

Row	Orders that fit all earlier rows and this row's clue
1	
2	
3	
4	

Problem 12: Compare this city with your five-clue city. Record how many clues each puzzle used and whether the completed cities agree.

My comparison:

Problem 1: Use R and B counters; repeats are allowed. A test scores one point for each matching position. Say only the total score. Use four positions. Hide a secret and make tests until you can identify it. Keep every test and score visible; add rows on scrap paper if needed. Naming a determined secret costs no extra test. Reveal and check your record.

Test	Code tested	Score
1		
2		
3		
4		
5		

Problem 2: Build each visible secret below. Test RRRR, then BRRR. Only the first test position changes. Record the scores and their difference.

Secret	RRRR score	BRRR score	Change
RBRB			
BRBR			
RRRR			
BBBB			

Problem 3: Keep a visible secret while you compare RRRR with each of BRRR, RBRR, and RRBR. Each comparison changes one position from the RRRR starting test. Record the scores for all four secrets.

Secret	RRRR	BRRR	RBRR	RRBR
RBRB				
BRBR				
RBBB				
RRRR				

Problem 4: Try those four tests on a hidden secret. Use each change from the RRRR score to find positions 1, 2, and 3. Count the Rs you have found. Subtract that count from the RRRR score to decide position 4. Record your answer, then reveal and check.

Test	Code tested	Score
1		
2		
3		
4		

My four positions and the baseline R count:

Problem 5: The first test RRRR scored 2. Build the six possible secrets below. For each proposed second test, record its score against each secret. In each row, circle the largest group of equal scores.

Test	BBRR	BRBR	BRRB	RBBR	RBRB	RRBB
BRRR						
BBRR						
BBBR						
BBBB						

Problem 6: For the second test BRRR, keep the secrets that would score 3. For the second test BBRR, keep the secrets that would score 2. Copy each group below. Try one final test against every secret in its group; record its scores.

BRRR answered 3: candidates, final test, and scores:

BBRR answered 2: candidates, final test, and scores:

Problem 7: Keep the three candidates BBRR, BRBR, and BRRB. Try the final tests below. Circle repeated scores within each row. Add a test of your own.

Final test	BBRR	BRBR	BRRB
RRRR			
RBRR			
RBBR			

Problem 8: Keep the four candidates BRBR, BRRB, RBBR, and RBRB. Try these final tests and circle repeated scores. Add a test of your own.

Final test	BRBR	BRRB	RBBR	RBRB
RRRR				
BRRR				
BBRR				

Problem 9: Take the first test BRBR and a secret of your choice. Swap R with B at positions 1 and 3 in both words, turning the test into RRRR. Record the two secret words and both scores. Repeat with a different secret.

Original secret	Score for BRBR	Changed secret	Score for RRRR

Problem 1: Two players alternate taking 1, 3, or 4 counters. Taking 2 is not allowed. Never take more than remain. Taking the last counter wins. Play from 8, 10, and 14. Record your moves. Swap who starts for a second attempt at a pile that surprised you.

Starting pile	First move	Next moves, in order	Winner: first or second
8			
10			
14			

Problem 2: W means the next player can force a win; L means every legal move gives the opponent a winning strategy. Zero is L. Build small piles and work upward: mark W if a legal move reaches L, and L if every legal move reaches W. Record the next piles as well as the labels.

Pile	Legal next piles	W/L
0	none	L
1		
2		
3		
4		
5		
6		
7		

Problem 3: Continue using moves 1, 3, and 4. Fill the chart in order. In each cell write L, or write W and a winning move: W1 means W with winning move 1. Check all legal next piles before marking L. Compare the rows after filling at least two of them.

Pile	0	1	2	3	4	5	6
Label/move							
Pile	7	8	9	10	11	12	13
Label/move							
Pile	14	15	16	17	18	19	20
Label/move							
Pile	21	22	23	24	25	26	27
Label/move							

Problem 4: Compare these four piles. Record every legal next pile and its label. Circle the starting piles that have no move to L.

Pile	Take 1: next pile/label	Take 3: next pile/label	Take 4: next pile/label
14			
16			
15			
17			

The repeating pattern I see:

Problem 5: Arrange 14, 16, 15, and 17 counters in whole groups of 7 with a smaller leftover group. Record the leftovers. Match each pile to its label in Problem 4.

Pile	Whole groups of 7	Leftover counters	W/L
14			
16			
15			
17			

Problem 6: Use your chart to choose a winning move for each leftover count marked W. Record the leftover count after the move. For counts marked L, list every leftover count an allowed move can reach; use piles 7 and 9 so that all three moves are possible.

Leftover count	W/L	Move to L, or all moves from L	New leftover count(s)
0			
1			
2			
3			
4			
5			
6			

Problem 7: Build 24 counters. Choose a first move using your table, then play. After each of your turns record the leftover count when grouping the remaining pile in sevens.

First move and later leftover counts:

Problem 8: Change the allowed moves to 1, 3, and 5. Play from 6 and from 7, then swap who starts. Build smaller piles to check a proposed strategy. Record which piles you want to hand the other player.

My games and the piles I want to leave:

Problem 9: Return to moves 1, 3, and 4. Copy the labels for piles 0–3 into the first row below and 7–10 into the second. These are four-label windows. To find the next label, inspect the first, second, and fourth cells: mark the next W if any is L, and L if all three are W. Record the next label, then slide the window one cell right.

Window starts at	Four labels	Next label	Window after sliding
0			
7			
1			
8			

Problem 10: Find another matching pair of four-label windows in your chart. Copy both windows, extend them by two labels, and compare the results.

My matching windows and their next two labels:

Problem 1: Build three piles. Take turns removing one or more counters from exactly one pile. Whoever takes the last counter wins. Play (1,2,3), then (1,2,4). Change who starts and play each again. Use the spaces for your piles.

Start	Who won?	One first move and reply
(1,2,3)		
(1,2,3)		
(1,2,4)		
(1,2,4)		

Problem 2: Replay one game. Choose a different first move and record the reply. Circle a start you would prefer to give your opponent.

Problem 3: Test these starts with counters. For each, try to leave two equal piles and one empty pile in a single move. Write a move if you find one, or record what you tried.

Start	Total	Move or attempts
(1,1,1)		
(1,1,2)		
(1,2,3)		
(2,4,6)		

Problem 4: Rebuild (1,2,3) after each trial. Your partner chooses each first move in this table. Find a reply leaving two equal piles and one empty pile; write the result.

Partner leaves	My reply leaves
(0,2,3)	
(1,0,3)	
(1,1,3)	
(1,2,0)	
(1,2,1)	
(1,2,2)	

Problem 5: Group each pile into bundles of 4, 2, and 1, using each size at most once. Record 1 if you use that bundle, 0 if you do not. The first row shows 5 counters grouped as $4+1$. Make and record the other piles with counters.

Pile	4	2	1	Bundle sum
5	1	0	1	$4+1$
1				
2				
3				
4				
6				
7				

Problem 6: Represent these two games using the same bundle records. Count the 1s in each column. Circle every odd count. Call a game balanced if all three counts are even; zero is even.

Pile	4	2	1	Pile	4	2	1
1				1			
2				2			
3				4			
Count				Count			

Problem 7: For (1,2,4), reduce just one pile so all column counts become even. Make that move with ordinary counters. Record the new piles.

Problem 8: For each start, write each pile’s 4/2/1 record. If all column counts are even, write “balanced.” Otherwise shrink one pile to make them even and record the new piles. Make each move with counters.

Start / pile	4	2	1	Move or balanced
2				
4				
6				
3				
4				
5				
1				
5				
7				
3				
5				
6				

Problem 9: Choose one balanced start above. Let your partner make a move. Record the new piles, then find and play a move that restores balance. Repeat with a different partner move.

Partner leaves	My reply leaves

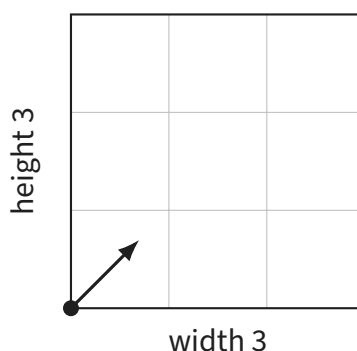
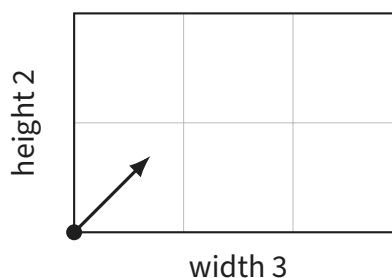
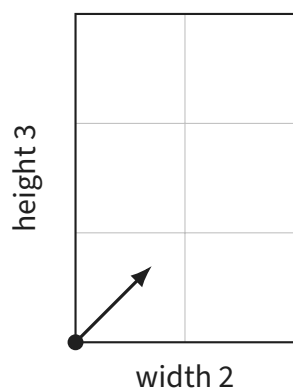
Problem 10: Try this way to repair odd columns. Find the largest bundle column with an odd count. Choose a pile using that bundle. Keep its larger bundles, remove this bundle, and choose its smaller bundles to make every column even. Try (4,6,7), then (7,10,12). Record the starting bundles and the replacement pile. Check that it is smaller, then make the move with counters.

Pile	8	4	2	1	Replacement pile
4					
6					
7					
7					
10					
12					

Problem 11: Choose a third pile that balances (2,5,____), then (3,6,____). Build both games and check their columns.

Problem 12: Choose two new piles and make a third that balances them. Give the game to a partner, then restore balance after their move. Record your start and one pair of moves.

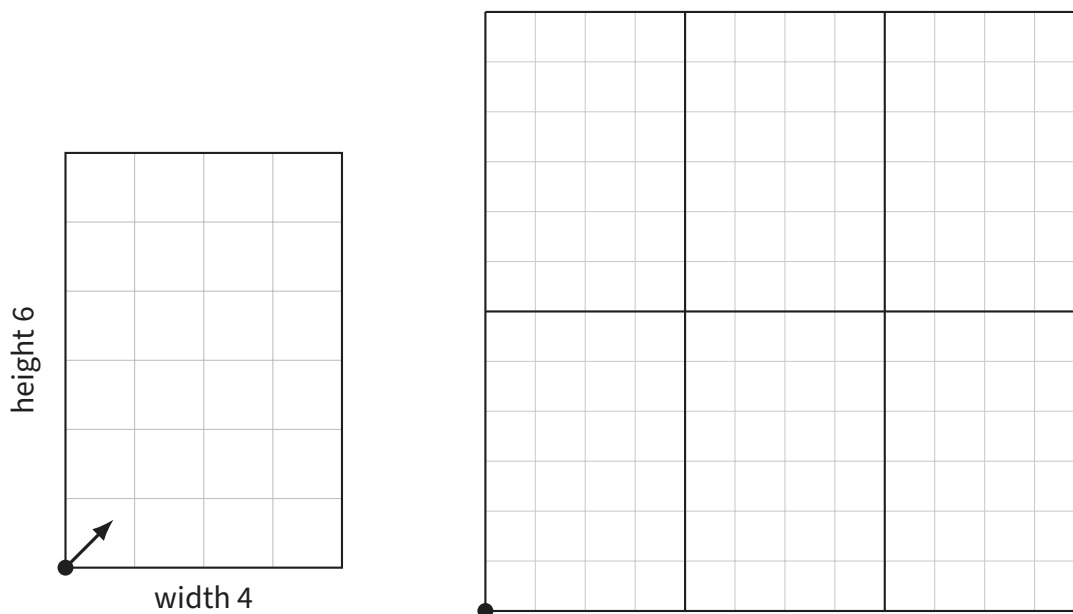
Problem 1: Start at each black dot with slope 1: one unit across for each unit up. Bounce at walls, reversing only the direction blocked by the wall. Stop at the first corner. Trace each path, mark wall bounces with dots, and circle the endpoint. Crossings of old path lines are not walls.



Problem 2: Record the ending corner and number of bounces for each table. Do not count the launch or final corner as bounces. Compare the two non-square tables.

Width $\times$ height	Corner	Bounces
2×3		
3×2		
3×3		

Problem 3: Trace the 4×6 table on the left. On the right are mirror copies of the same room. Draw a straight diagonal from the black dot to the first intersection of a vertical and horizontal room wall. Number the wall crossings before that intersection and match them to bounces on the left.



Problem 4: List the horizontal distances where the straight path crosses a vertical room wall. List the vertical distances where it crosses a horizontal room wall. Circle the first number appearing in both lists. The two distances agree along this diagonal.

Problem 5: Use your circled distance to count whole room widths and heights. Move your finger back and forth across the original width for each whole width, then up and down for each whole height. Record the side reached in each direction and the resulting corner.

Problem 6: Use wall-distance lists to find the first common distance L for each table. Divide it by the width to count whole widths, and by the height to count whole heights. Record the folded corner. Odd whole-width counts end right; even counts end left. Odd whole-height counts end top; even counts end bottom.

Table	L	Widths	Heights	Corner
4×6				
6×9				
6×10				
8×14				

Problem 7: Compare 4×6 with 6×9 . Draw or record how the table and unfolded distances changed. Then compare 4×6 with 6×10 , whose sides are also both even. Circle the two tables with different ending corners.

Problem 8: On the 4×6 copied rooms, extend your imaginary diagonal to distance 24. It would cross 6 whole widths and 4 whole heights. Record the corner those counts predict, then mark the earlier corner where this game actually stops.

Problem 9: For each table, list the distances of vertical and horizontal wall crossings before its first corner. Count all these crossings. Compare your counts with your earlier traced paths.

Table	Vertical walls	Horizontal walls	Bounces
2×3			
3×3			
4×6			
6×10			

Problem 10: Choose a table from Problem 9. Draw its copied rooms or record both wall lists. Circle its final corner separately. Count how many vertical crossings are left when the final corner is removed; then count horizontal crossings. Write a rule using whole-width count m and whole-height count n .

Problem 11: Make a scaled copy of your chosen rectangle by doubling both sides. Predict its corner and bounce count. Check using wall distances and record your result.

Problem 12: Try these three rectangles. Trace their paths and record corners and bounce counts. All have width plus height equal to 6. Circle those reaching top-right after exactly four bounces.

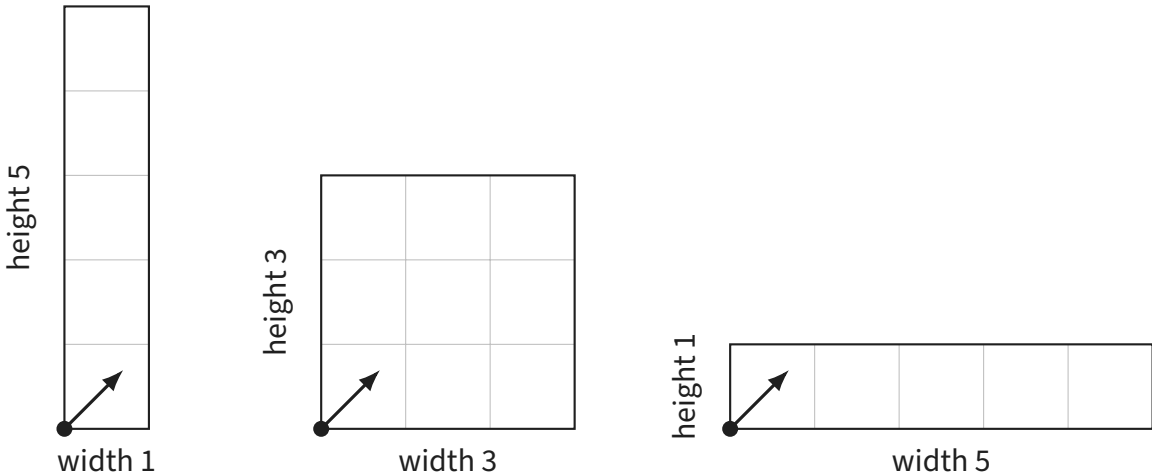
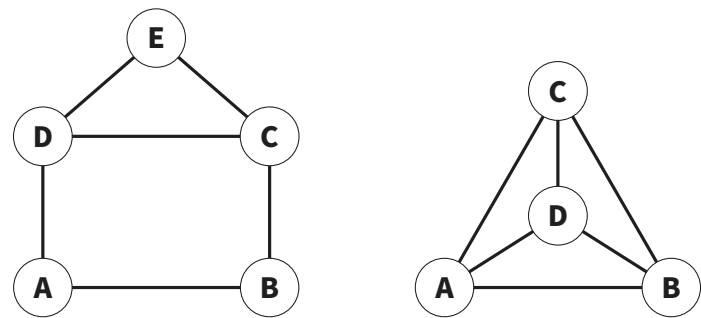


Table	Corner	Bounces
1×5		
3×3		
5×1		

Problem 13: Make a different-size rectangle ending top-right after four bounces. Record its size, first common distance, and whole-room counts.

Problem 14: Try to make a table ending top-right after three bounces. Use the whole-room counts to list your candidates. Record which corner each candidate reaches, or which earlier corner stops the path.

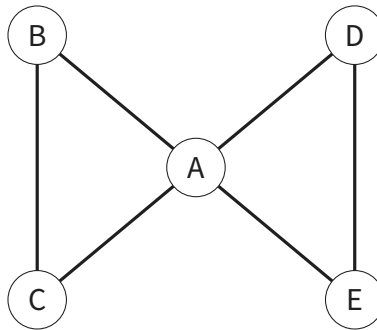
Problem 1: Try to use every line once without jumping. Revisiting a dot is allowed. Mark used lines with counters. Try the house, then the triangle with center D. Choose two different starting dots on each. Record a complete route, or your attempted route and unused lines.



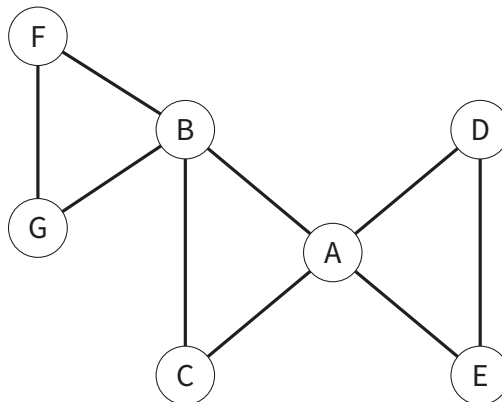
Map	Start	Route	Unused lines
House			
House			
Triangle			
Triangle			

Problem 2: Count each dot’s incident lines, called its degree. Circle the odd-degree dots. Choose your complete house route and pair the arrival/departure lines at each internal visit. Record the dots with one unpaired line.

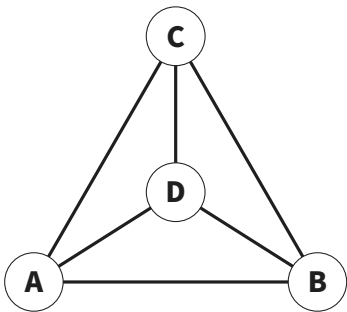
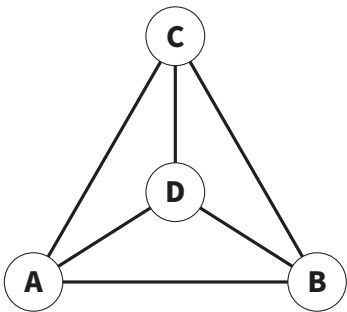
Problem 3: Walk A–B–C–A and mark those roads used. Walk the remaining loop A–D–E–A. Write each loop on a paper strip. Join the strips at A to make one route using all six roads. Record and trace the joined route.



Problem 4: The map below adds a loop at B. Start with your route from Problem 3, then insert B–F–G–B at a visit to B. Record the joined route and check every road is used once. Count the degree at B before and after the new loop.



Problem 5: Now travel every road at least once and return to A. Roads may be repeated; each traversal costs 1 step. Find a route on the left map and record its visited dots. Try a different route on the right map with fewer steps, or match your best count. Mark each repeated road with an extra pencil line.

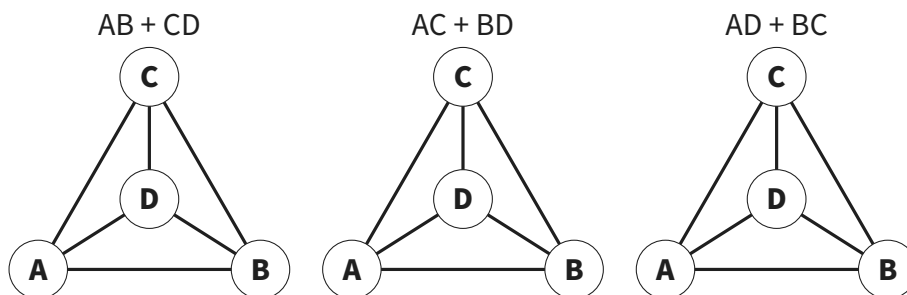
Trial	Route	Steps
1		
2		

Problem 6: Count how many times your best route uses each road. Record the original six crossings separately from the repeats. At each dot, count all arrival and departure crossings, including repetitions. Circle any odd total.

Road	AB	AC	AD	BC	BD	CD
Times used						

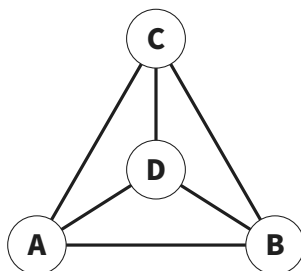
Problem 7: Add one extra copy of a road to a clean map. Count how many odd degrees change to even. Then add one more road copy that makes all degrees even. Record the two roads.

Problem 8: Pair the four odd dots in each way shown. Draw a second copy of each named road on its map. Then find a closed route using every original and added road once. Record its length.

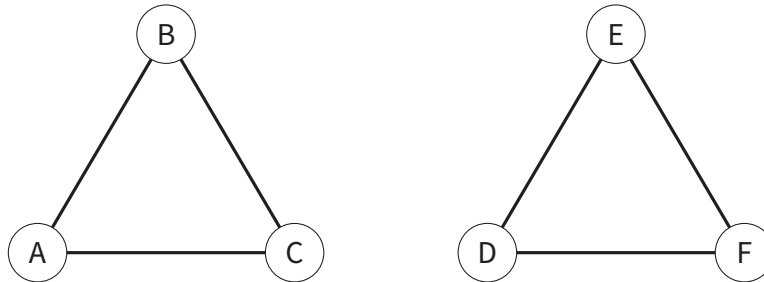


Pairing	Closed route	Length
AB + CD		
AC + BD		
AD + BC		

Problem 9: Now the truck may finish at a different dot. Add just one road copy to the original map below, then find a route using every original and added road once. Record its endpoints and length.



Problem 10: Add three roads between these two triangles so the map is connected and all six dots have odd degree. Curve roads around other dots; a crossing without a dot is not a junction. Count the degrees. Then choose three roads to repeat so every degree becomes even. Record a closed delivery route and its length.



Problem 11: This map also has six odd dots. Start at A, visit every road, and return to A. Mark each repeated road and record your route. Compare how many extra crossings the two maps need.

